FIELD CRAFT

BATTLE CRAFT

2

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FIELD CRAFT AND BATTLE CRAFT (SD/SW)

CHAPTER FCBC-I: INTRODUCTION TO FIELD CRAFT AND BATTLE CRAFT

"If you know heaven and earth, you may make your victory complete."



TEACHING INSTRUCTIONS

Period	**:**	**03 (120 Mins)**
Type	**:**	**Lecture and Practice**
Year	**:**	**1st Year SD/SW**
Conducting Officer	**:**	**Permanent Instructor**

Training Aids: Class Room, Computer with OHP & Screen, Pointer Staff, Presentation, Precis Lesson plan, Board & markers and Open training area for practical classes/ Practice.

Time Plan

➢ **Introduction**	**:**	**02 Mins**
➢ **Part I**	**:**	**10 Mins**
➢ **Part II**	**:**	**28 Mins**
➢ **Part III**	**:**	**17 Mins**
➢ **Part IV**	**:**	**20 Mins**
➢ **Conclusion**	**:**	**03 Mins**
➢ **Practice**	**:**	**40 Mins**

INTRODUCTION

1. In view of the multifaceted expectations and contribution that is expected/ assigned to cadets during peace, war or emergency situation, Battle Craft (BC) & Field Craft (FC) has a non-negotiable role in explaining nature of ground, means to advance towards target and finally accomplishing the allotted task. During war like situation the importance of BC/FC increases manifolds, because locating the target correctly, informing the locations to the forward elements and engaging the targets becomes of paramount importance. Similarly locating people during natural calamities and selecting the most favorable route of saving them also demands correct identification of landmarks.

INTERESTING FACTS

- **The art of field craft and battle craft dates back to the ancient times when it was used for food and survival.**

- **Pathani Samanta the great astronomer and mathematician who hails from Odisha could measure distance using bamboo pipe & traditional instruments. He could also measure the distance between earth & Sun & other stars by using one bamboo pipe & two wooden sticks.**



PREVIEW

The lecture will be conducted in following parts:-

- **Part I: Terminologies used.**
- **Part II: Field Craft, Battle Craft & Practice.**
- **Part III: Method of Judging Distance & Practice.**
- **Part IV: Types of Ground, Identification of Landmarks & Practice.**

LEARNING OBJECTIVES

- **Familiarizing cadets with FC and BC.**
- **Role of FC, BC.**
- **Various subjects that come under FC, BC.**
- **Means to Judging Distance.**
- **Methods of Judging Distance.**
- **Practice on Ground.**

PART I: TERMINOLOGIES USED

2. **Terminologies used**. In this part the cadets shall study the various terms and definitions that will be used during the course of the chapter.

(a) **Landmarks**. A natural or artificial object on the ground which is used for the verbal orders irrespective of whether its present on map or not.

(b) **Reference Point**. An important and unmistakable object, with the help of which you can indicate other landmarks and targets. A reference point should be specific and if more than one degree in expansion, corners can be taken as reference.

(c) **Target**. It is an object having technical significance which is indicated with a view to bring down fire on it. In case of natural calamities this will be the place where we are expected to give our support.

(d) **Ground**. It is the area available for accomplishing the main task.

(e) **GRAD**. To be explained through examples:

G	**Group**	The affected team for the task
R	**Range**	Distance of Object from own location
A	**Aid**	Assistance taken to describe the target
D	**Description**	Description of the target

(f) **Field Signals**. Methods used by the team commander to explain the target & methods to accomplish task allotted to his team.

(g) **Formation**. The ways and means adopted by an Infantry section to advance towards the target.

(h) **Fire Control Orders**. Orders given by the team commander to engage designated targets.

(i) **Camouflage & Concealment**. Act of hiding & surprising the enemy and engaging him without disclosing own whereabouts.

PART II: FC & BC

3. Field Craft (FC) is the skill of using the ground and the weapon to the best of one's own advantage. Whereas Battle Craft (BC) is a set of drills which are essential for the conduct of successful tactical movement or operations in the (battlefield). So whilst FC can be practiced by a trained person singularly or convectively, BC is a collective action by a group of trained personnel like a section or a platoon.

4. FC is an essential stepping stone of soldiering. It involves mastering multiple techniques and processes such as camouflage, concealment, estimating distance, etc., so that a soldier can exploit any situation to his advantage. These are:-

(a) **Visual Training**. Developing the ability to quickly and accurately identify targets.

(b) **Camouflage and Concealment**. Using natural and artificial materials to blend into the environment and avoid detection.

(c) **Judging Distance**. Accurately estimating the distance to targets for effective engagement.

(d) **Movement Techniques**. Utilizing terrain and cover to move safely with or without arms.

(e) **Field Signals**. Using hand and arm signals, with or without sound, to communicate with other soldiers.

(f) **Recognition and Description of Targets**. After acquiring visual training skills, soldiers learn to recognise targets both on the ground and on maps, and to accurately describe them to others. This ensures that targets are clearly understood without error or confusion.

5. **Movement Techniques**. During war it is imperative for everyone to advance towards the target/object without being detected. Various methods of advancing towards the object can be crawling, rolling, side rolling with or without arms. Same will be demonstrated during practice.

MOVEMENT WITHOUT ARMS	MOVEMENT WITH ARMS
The Monkey Run	The Ghost Walk
The Leopard crawl	The Monkey Run
The Walk	The Leopard crawl
The Roll	The Moderate Walk
Crawl through abdomen	The Roll
The Kitten Crawl	
The Cat Walk	

THE CAT WALK



THE GHOST WALK



THE ROLL WITH WEAPON



THE ROLL WITHOUT WEAPON

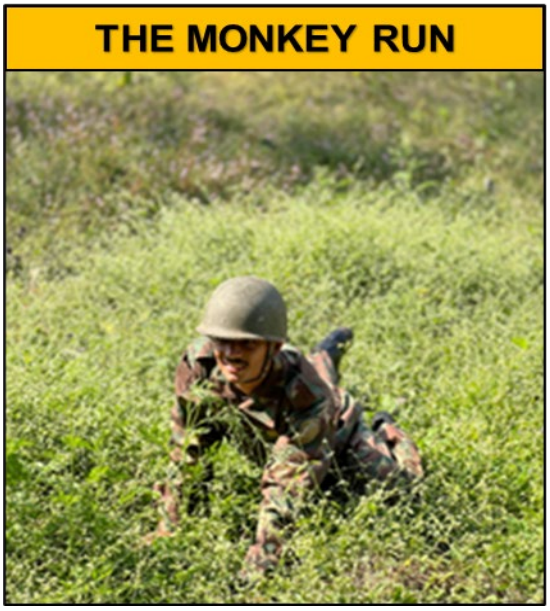


CRAWL THROUGH ABDOMEN



THE LEOPARD CRAWL (A)







6. **Battle Craft**. Battlecraft set of drills which are essential for conduct of successful operations in the battlefield. These battle drills are very useful in tackling minor tactical problems. They save time, ensure rapid action and avoid confusion. Knowledge of field signals and section and platoon formations, however is essential in the execution of various battle drills. Battle craft includes the following subjects:-

(a) **Field Signals**. Sometimes in the battle field, there is a need for a complete silence. Then “signals” are applied in place of verbal communications.

(b) **Section Formations**. A section is a small military unit consisting of 10 personnel. Formations used when Section comes in contact with the enemy is called Section Formation.

(c) **Fire Control Orders**. Fire Control orders are a set of instructions given to soldiers to help them identify and shoot a target. Fire control orders are given in a specific sequence and soldiers should know how to respond to them. It is a specific command given by a fire unit leader to direct their troops on exactly where and how to engage a target, including details like range, type of fire, and target indication allowing for precise and coordinated firepower.

(d) **Fire and Move**. Fire and move, also known as fire and maneuver, is a military tactic that involves movement of troops while using fire to suppress the enemy. It is used to advance on enemy positions or disengage from them.

(e) **Fire Discipline**. Fire Discipline refers to the established system of words, phrases, rules and conventions used in military and emergency response contexts to ensure precise, efficient, and timely action. In military context, Fire Discipline is a standardized vocabulary and set of conventions for directing fire of artillery guns and other weaponry. The goal of same is to ensure that commands are understood with precision, eliminating ambiguity and confusion, and leading to immediate and accurate action at the guns. The system includes initial calls for fire, target

descriptions, adjustment orders and commands for fire for effect, all delivered in a specific sequence. Fire Discipline assures orderly operations, maximises safety and leads to successful outcome on fireground.

(f) **Section Battle Drills**. Section battle drill is a military drill that involves preparation, reacting to enemy fire, and regrouping.

PART III: JUDGING DISTANCE & METHODS

7. Judging the correct distance helps to effectively engage any target during war. Major factors that affect Judging Distance are:-

(a) Nature of ground.
(b) Experience of observer.
(c) Ambient light.
(d) Intervisibility of objects.
(e) Weather conditions.



8. Accurate fire with any weapon depends on the correct judging of distance. Although a soldier is not normally required to open fire at range over 300 meters, he must be able to judge distance up to about 1000 meters, so that it distance enables the following:-

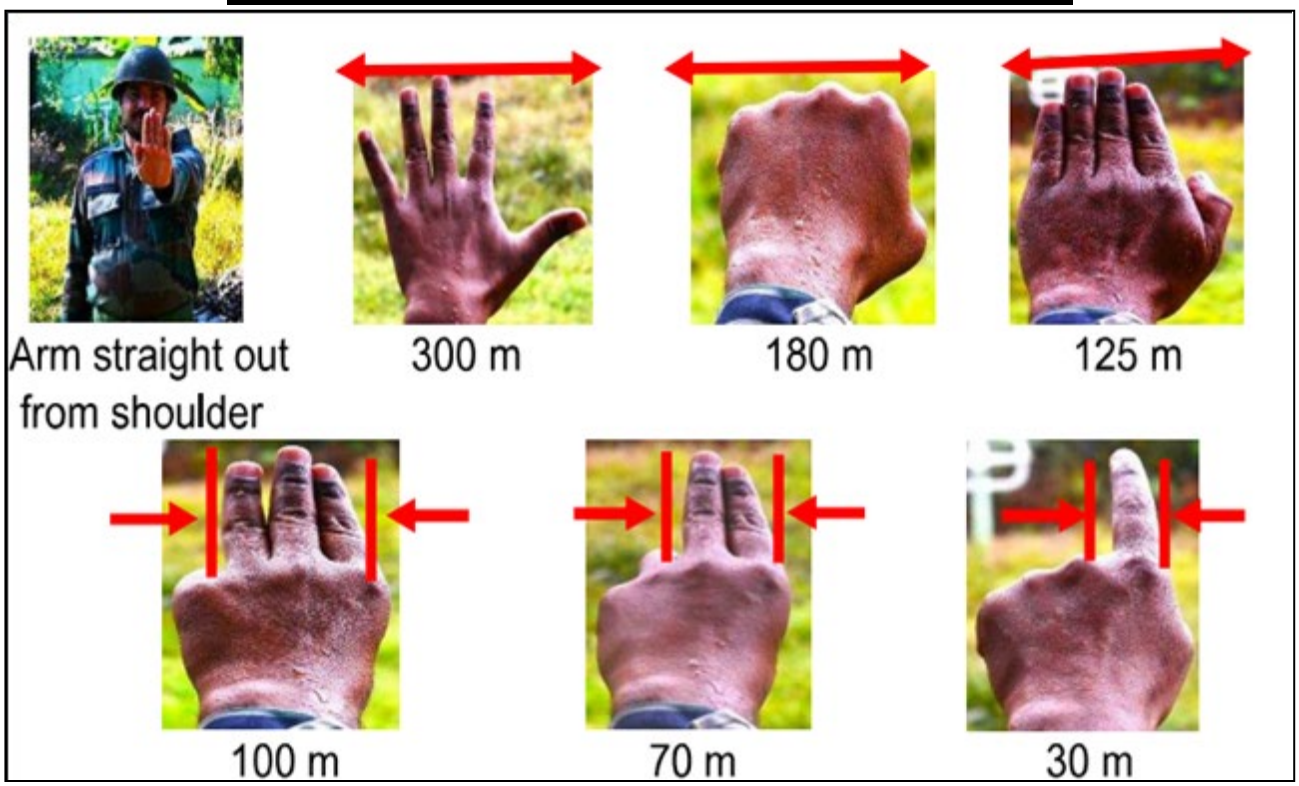
(a) To know when to open fire.

(b) To know which weapon to be used.

(c) Can indicate targets to other men in his section.

(d) Pass back accurate information when acting as an observer.

JUDGING DISTANCE BY HAND AT ONE KM (1000 M)



Methods of Judging Distance

9. The following are the six methods of judging distance:

(a) Unit of Measure.

(b) Appearance method.

(c) Section average.

(d) Key range.

(e) Halving.

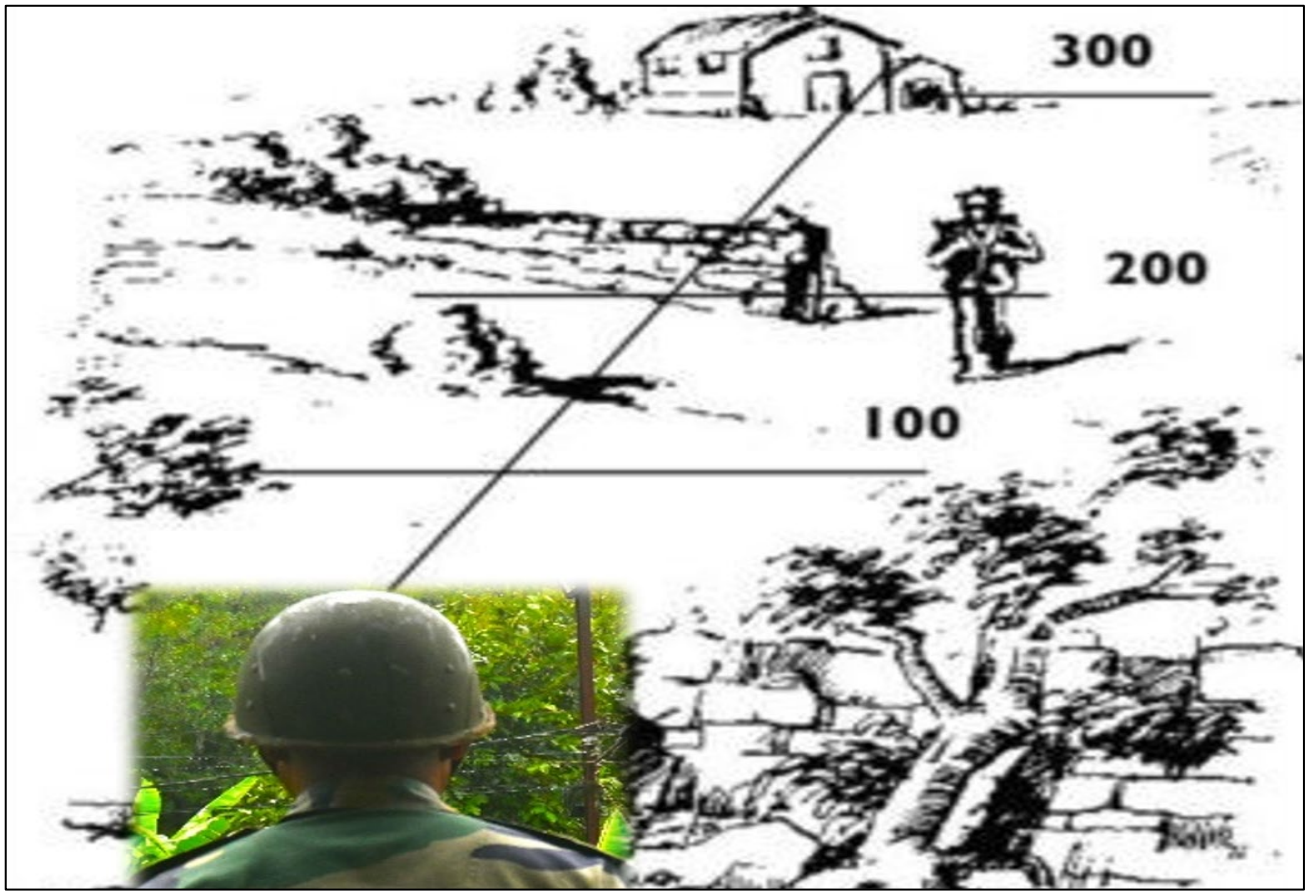
(f) Bracketing.

10. **Unit of Measure Method**. This method is also termed as the 100 meters method. The unit of measure chosen is normally 100 meters and therefore one should form a good idea of 100 meters distance on the ground. The length of a hockey field is the best yard stick for this purpose.

11. The distance of a given object will be a multiple of the imaginary unit of 100 meters, as placed between the observer and the object.

12. This method is not accurate above 100 meters and is of little use if there is dead ground between the observer and the object.

UNIT OF MEASURE METHOD`



13. **Appearance Method**. The distance can be judged by noting the detailed appearance of man at various ranges. This is the best method under service conditions. The following is a guideline to judging distance by this method:-

(a) At 200 meters, all parts of the body are distinct.

(b) At 250 meters, blade of the foresight covers a kneeling man.

(c) At 300 meters, the face becomes blurred.

(d) At 400 meters, the body remains same in shape but face is difficult to distinguish. Blade of the foresight covers a standing man.

(e) At 500 meters, body appears to taper slightly from the shoulder but movement of limbs can still be seen.

(f) At 600 meters, head appears as a dot. Details are not visible and body tapers from shoulders downwards noticeably.



APPEARANCE METHOD

At 100 m – clear in all details.





At 200 m – clear in all details, color of skin and equipment identifiable.





At 300 m – clear body outline, face color visible, remaining details blurred.

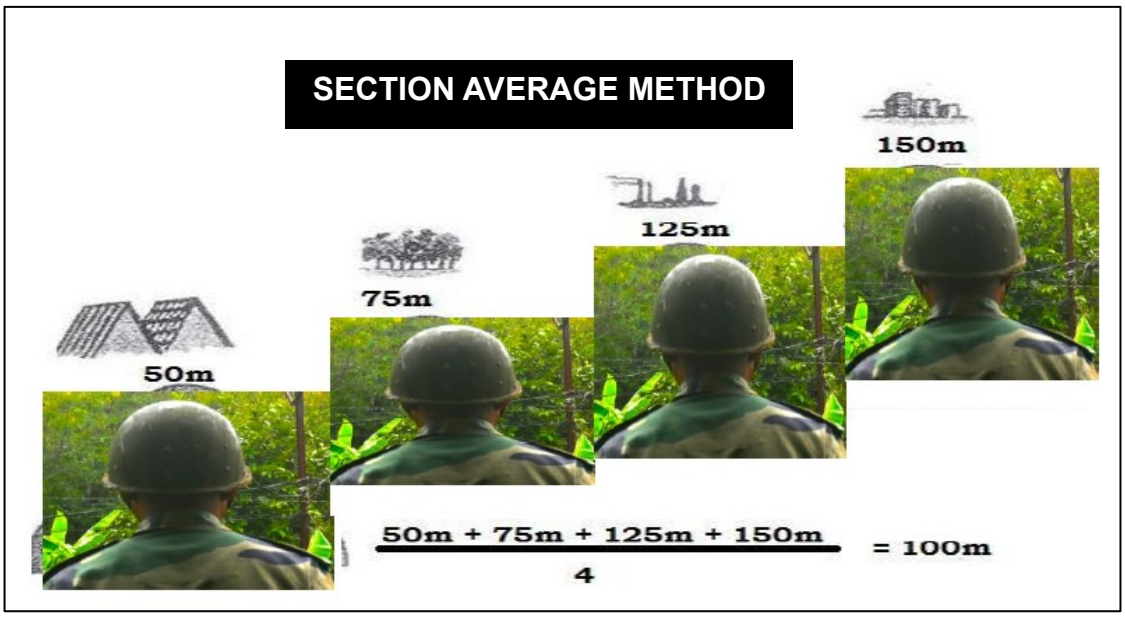


14. **<u>Section Average Method</u>**. Each man in the section is asked to judge the distance of a given object. The average of the answers given by the whole section is then accepted as the distance of the target from the observer. Here caution must be exercised in the estimation of a few who may foolishly overestimate/ underestimate the distance. This method may be resorted to under the following circumstances:-

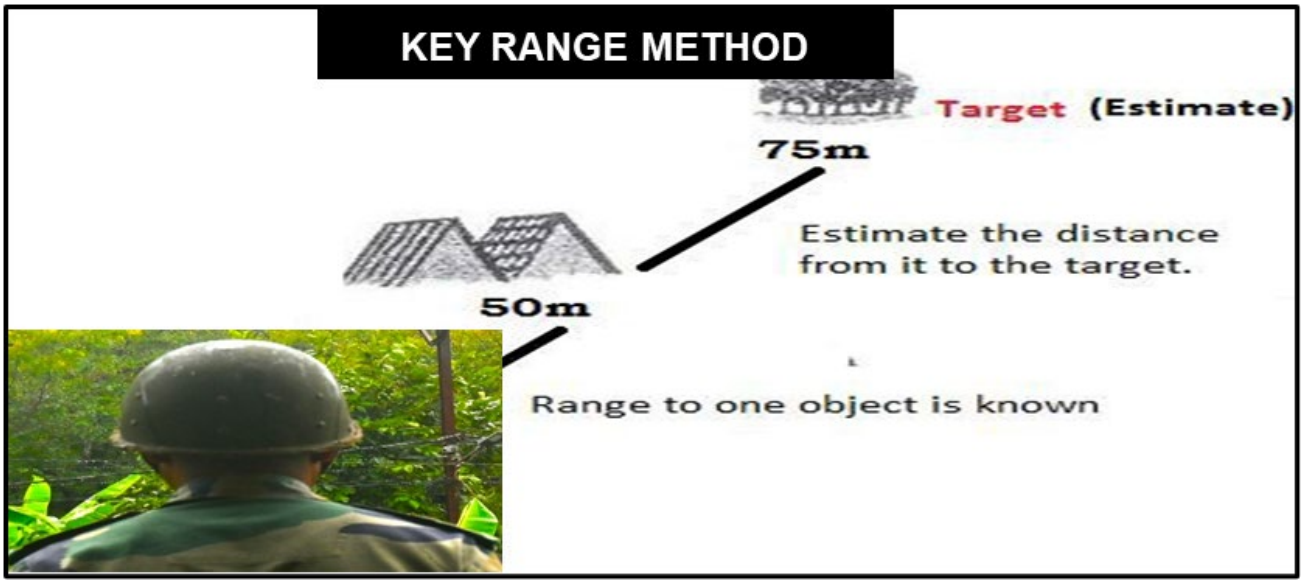
(a) Ample time is available.

(b) Judging of distance is made difficult by mist or darkness.

(c) Judging of a long distance is involved e.g. beyond 400 meters.



15. **Key Range Method**. If the range of the certain object is known, distance to other objects can be found in relation to the known range. This method is called 'Key Range' method.



16. **Halving Method**. Halving Method entails that an object is selected halfway between the observer and the target, the distance to the selected object is judged and doubled to get the distance to the target.

17. **Bracketing Method**. The observer works out the maximum and the minimum possible distance of the object and then accept the mean as the distance e.g. maximum possible distance 1000 meters, minimum possible distance 500 meters therefore estimated range is 750 meters. The greater the range, wider the bracket. In no case the bracket should be less than 300 meters.

DID YOU KNOW?

- Field Craft and battle Craft procedures are enormously affected by the terrain and weather conditions.
- Correct Judging of distance assists you in a big way in your day- to – day activities as well.
- Learning to trust your compass and night navigational aids also play a vital role in FC & BC.
- Distance up to 300 m is called near distance, Distance between 300 - 500 m is called middle distance and distance beyond 500 m is called far distance.

Practical Hints

18. **During Day**. During the day judging distance will depend on the inter visibility of observer and the object. Therefore, whilst judging the distance, the aspect which leads to over estimation or under estimation of distances should be kept in mind. A combination of 'Key Range method' with another suitable method should be used in arriving at the best

answer. In addition to this, comparative appearance of a known object with generally standard dimensions like a stationary car, a regular sized door, corridor of a building at various distances can facilitate in judging the distance of the object. For distances of about 300m or lesser, Thumb method can also be used. This works on the principle that the approximate ratio of the arm length and interpupillary distance (distance between eyes) when the arm is outstretched is 10 times. The process involves extending your arm, closing one eye, aligning your thumb with the object. Now switch eyes, you would notice the apparent movement of the thumb. Multiplying the apparent shift of thumb from the edge of the object by 10 will give you the actual distance.

19. **During Night**. During night and hours of poor visibility, judging distances with naked eyes becomes a very challenging process. At night the visibility will be governed by ambient light and phases of moon, as well as topography, viz. on a snow covered or plain desert, visibility is significant during full moon phase at night. Apart from this, as surroundings become darker, the distances start to appear bigger. An observer can keep in mind certain very large objects with identifiable silhouettes, distance of which is known, like a multi storied building or a tree, and use them to identify the distance of other objects at night. A sound of an object can also give a perception of the approximate distance of the object. An observer must keep all the above mentioned factors in mind, and should commence his or her observations at night by carrying out low light adaptation drill for about 20–30 minutes in which the eyes get accustomed to dusk/night situation.

20. Distances are wrongly estimated under following conditions:-

(a) **Distance is over estimated when**.

(i) Light is bad.

(ii) The sun is in the observer's eye.

(iii) The object is small in relation to its surroundings.

(iv) Looking through valley of narrow land e.g. street.

(v) Lying down.

(b) **Distance is under estimated when**.

(i) The light is bright or the sun is shining from behind the observer.

(ii) The object is large in relation to its surrounding.

(iii) There is some dead ground between observer and the object Looking uphill.

PART IV: TYPES OF GROUND & IDENTIFICATION OF LANDMARKS

21. A standard, quick and accurate procedure is necessary to enable commander to describe an area to his men quickly and the men to understand it correctly and vice versa.

Definition

22. There are four types of ground:-

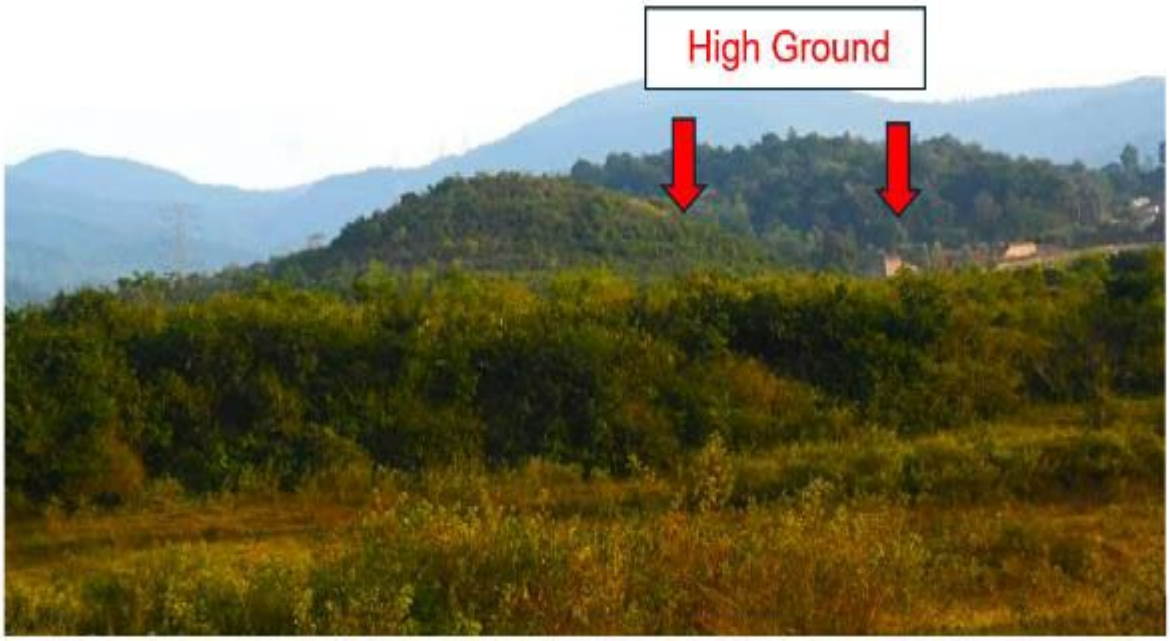
(a) **Broken Ground**. It is uneven and is generally interspersed with nullahs, bump sand field in the ground. It is suitable for move of infantry and hinders observation of activities.



(b) **Flat and Open Ground**. It is even ground with little cover eg. bushes, hedges and similar foliage. It is not suitable for move of Infantry by day.



(c) **<u>High Ground</u>**. Ground far above the general level of the area e.g. hill. It facilitates domination of area around it by observation or fire or both.



(d) **<u>Dead Ground</u>**. Ground that is hidden from an observer's view. It cannot be covered by flat trajectory weapons



23. The following points should be remembered about various types of ground:-

(a) Though an open ground is easy to travel, it is dangerous to do so in the vicinity of the enemy. Whether moving or taking fire position in an open area one is vulnerable to enemy from view and fire.

(b) Broken ground when correctly used affords protection from flat trajectory weapons. It does not afford cover from air or protection from high trajectory weapons.

(c) Dead ground does not afford cover from high trajectory weapons.

Procedure of Description of Ground

24. The normal method of scanning and describing ground is by dividing it as follows:-

(a) **Fore/Near Ground**. Any object / landmark up to distance of 300 meters is called Fore / Near Ground.

(b) **Middle Ground**. Any object / landmark between distance of 300 meters up to 500 meters is called Middle Ground.

(c) **Far/Distant Ground**. Any object /landmark beyond 500 meters is called Far/Distant ground.

25. After theoretical class, the cadets will be given exposure of subject practically outdoor. The practice will include the following:-

(a) Movement with & without arms.

(b) Various methods using arm to judge distance.

(c) Practice of using various methods of judging distance using natural resources.

(d) Practice of judging distance during night.

26. **Identification of Landmarks**. Points to keep in mind while identifying landmark are as follows:-

(a) It should be prominent.

(b) It should be static in nature.

27. The following points should be remembered about various types of ground:-

(a) Though an open ground is easy to travel, it is dangerous to do so in the vicinity of the enemy. Whether moving or taking fire position in an open area one invulnerable to enemy from view and fire.

(b) Broken ground when correctly used affords protection from flat trajectory weapons. It does not afford cover from air or protection from high trajectory weapons.

(c) Dead ground does not afford cover from high trajectory weapons.

PART V: PRACTICE

28. After theoretical class, the cadets will be given exposure of subject practically outdoor. The practice will include the following:-

(e) Movement with & without arms.

(f) Various methods using arm to judge distance.

(g) Practice of using various methods of judging distance using natural resources.

(h) Practice of judging distance during night.

CONCLUSION

29. To achieve success in war it is very important that the target is correctly understood alongwith mechanics of field and battle crafts. Knowledge of ground along with basic skills would be beneficial to cadets in all respect. Use of natural resources plays a vital role in effective execution of field craft and battle craft drills.

SUMMARY

- Field Craft include Visual Training, Recognition & Description of targets, camouflage & Concealment, Judging Distance, Move with and without Arms & fire Description and Control.
- Battle Craft include Field Signals, Section Formations, Fire Control Orders, Fire & Move and Section Battle Drills.
- There are six methods of Judging Distance.
- Ground plays a vital role in any operation of war & can be described in four major types.
- The procedure of describing ground is covered in three major phases.
- Accuracy in Judging Distance during day as well as night demands immense practice.
- Field Signals are effective ways of correctly identifying the target.
- Things are seen because of Shape, Shine, Shadow, Silhouette, Speed, Smell & Surface.

ASSESSMENT EXERCISE

Multiple Choice Questions and fill in the Blanks

Q1. In the _____ you judge the distance with the help of the blade of the foresight of your weapon.

(a) Rifle Method (b) Foresight Method
(c) Blade Method (d) Appearance Method

Q2. Which of these does not fall under the Field Craft.

(a) Section Formation (b) Visual Training
(c) Judging Distance (d) Fire Control

Q3. _____ are very useful in tackling minor tactical problems.

(a) Field Signals (b) Battle Drills
(c) Field Crafts (d) Section Formations

Q4. The art of using the ground and the weapon to the best of one's own advantage is called _____.

(a) Camouflage (b) Field Signal
(c) Field Crafts (d) Battle Craft

Q5. 'Unit of Measure' method is also called '_____ meters method'.

(a) 25 (b 150
(c) 50 (d) 100

Q6. Sequence of giving fire control order is _____, _____, _____ & _____.

(a) GRIT (b TRIG
(c) RGIT (d) TIRG

Q7. Section battle drill is conducted in _____ steps.

(a) 01 (b 02
(c) 03 (d) 04

Q8. Appearance method is one of the methods for _____.

(a) Fire Control (b Concealment
(c) Judging Distance (d) Battle Craft

Q9. There are _____ types of Ground.

(a) 03 (b) 04
(c) 05 (d) 06

Q10. Drill essential for conduct of successful operation is called _____.

(a) Field Craft (b) Battle Craft
(c) Battle Drill (d) Field Signal

Q11. Object/ Landmark identified as middle distance will be _____ metres away.

(a) 75 (b) 525
(c) 195 (d) 450

Q12. Distances are overestimated when:

(a) Looking uphill (b) Dead Ground between Observer & Object

(c) Light is bad (d) Light is bright

Q13. An individual should be able to judge distance to

(a) Know when to open fire (b) Know which weapon to use
(c) Both (a) and (b) (d) To carry out section drill

Q14. Ground that is hidden from observer's view is called _____.

Q15. Bracket should not be less than _____ meters.

Short Answer Questions

Q1. Explain GRAD.
Q2. Write the methods of Judging Distance.
Q3. What are the types of ground?
Q4. What is the sequence of giving control orders?
Q5. Section Battle Drill is conducted in how many and what steps?
Q6. What is the procedure for describing ground?
Q7. What does Bracketing method of judging distance means?

Long Answer Questions

Q1. Describe FC.
Q2. How do you recognize / describe a target?
Q3. What all subjects are include in FC? Explain in details.
Q4. What all subjects are include in BC? Explain in details.
Q5. Describe Battle Craft.
Q6. Differentiate between movement without arms and movement with arms.

FIELD CRAFT AND BATTLE CRAFT (SD/SW)

CHAPTER FCBC-II: INDICATION OF LANDMARKS AND TARGET

"Water shapes its course according to the nature of the ground over which it flows. The soldier works out his victory in relation to the foe whom he is facing."



TEACHING INSTRUCTIONS

Period	**:**	**05 (200 Mins)**
Type Year	**:**	**1st (02 periods), 2nd (02 periods) & 3rd Year (01 period) SD/SW**
Conducting Officer	**:**	**Permanent Instructor**

Training Aids: Class Room, Computer with OHP & Screen, Pointer Staff, Presentation, Preci Lesson plan, Board & Markers and Open training area for practical classes/ Practice.

Time Plan

➢ **Introduction**	**:**	**05 Mins**
➢ **Part I**	**:**	**05 Mins**
➢ **Part II**	**:**	**30 Mins**
➢ **Part III**	**:**	**35 Mins**
➢ **Conclusion**	**:**	**05 Mins**
➢ **Practice**	**:**	**120 Min (80 Mins for 2nd Yr & 40 Mins for 3rd Yr)**

INTRODUCTION

1. Landmarks and other objects on the ground or a battle field may be either unclear due to climatic conditions or other reasons. Every effort should, therefore be made to indicate their location and size carefully and accurately. To ensure quick and accurate indication by commanders and recognition by individual soldiers a standard procedure has been laid down in the Army. Even the aids to be used for indicating difficult targets have been laid down.

PREVIEW

The lecture will be conducted in following Parts:-

- **Part I: Definitions**.
- **Part II: Sequence of Description**.
- **Part III: Methods of Indication of Target**.
- **Part IV: Practice**.



INTERESTING FACTS

- **Correct Indication of Landmarks can be handy during rescue operations as well**.
- **Acquiring the target during any operations of war is directly proportional to indication of landmarks**.
- **Reference points make indication of landmarks easier**.
- **Procedures for indication of landmark varies with approximate distance of the object**.

LEARNING OBJECTIVES

- **Familiarizationwith Terminologies used and their definitions understanding**.
- **Sequence of description of Ground**.
- **Various methods used for identification & indication of targets**.

PART I: DEFINITIONS

2. **Landmarks**. An object which is important on the ground and is used in verbal orders to explain the ground in front.

3. **Target**. An object having a technical significance which is indicated with a view to bring down fire on it.

4. **Reference Point**. An important and unmistakable object, with the help of which you can indicate other landmarks or targets. A reference point should be specific.

5. Though an open ground is easy to travel, it is dangerous to do so in the locality of the enemy. Whether moving or taking fire position in an open area, one is exposed to enemy from view and fire. Broken ground when correctly used affords protection from flat path weapons. It does not afford cover from air or protection from high path weapons. Dead ground does not afford cover from high path weapons.

6. **Procedure of Description**. The normal method of scanning and describing ground is by dividing it as follows:-

(a) Distance up to 300 meters is called Fore Ground.



(b) Distance from 300 meters to 500 meters is called Middle Ground.



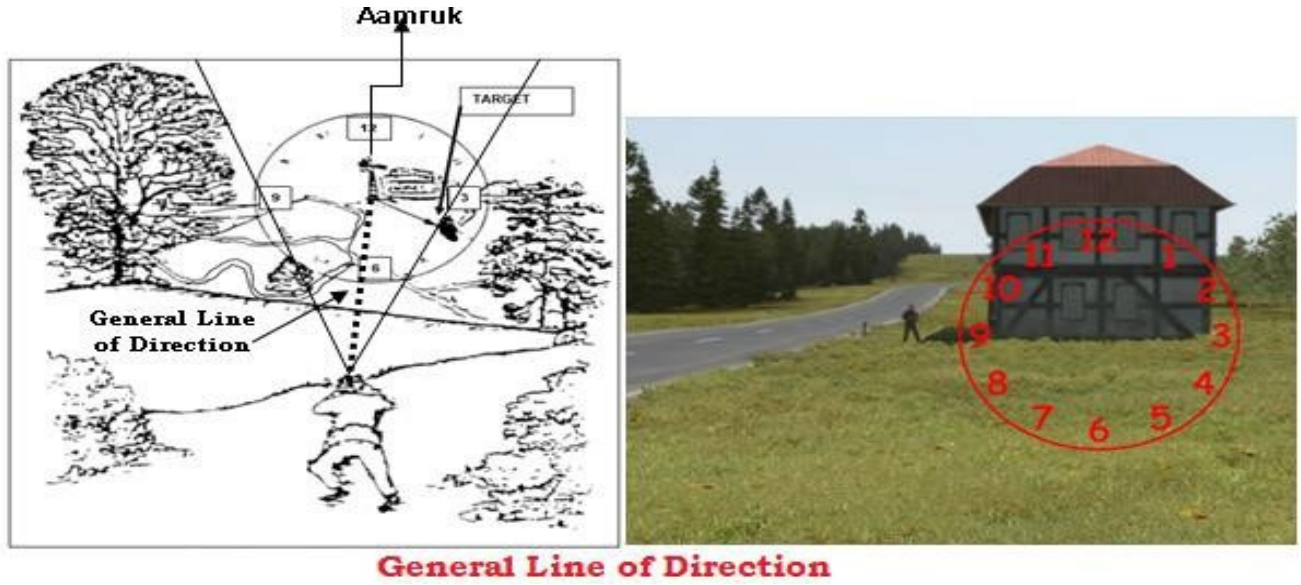
(c) Distance beyond 500 meters is called Distant Ground.



PART II: INDICATION OF LANDMARKS

7. **Sequence of description**. While describing the ground bounded by particular arc after giving the boundaries start from LEFT to RIGHT. If the ground all around is to be described, start after general line of direction to the right and finish at general line of direction by completing the indication all around.

8. **Identification of Landmarks**. Start by giving the general line of direction by pointing out a centrally located, if possible, prominent land mark, e.g. No 1 section 500 Red house, Section ka Madhya and line beyond General Line of Direction (GLD).



General Line of Direction

9. **Boundaries**. After giving general line of direction give LEFT and RIGHT boundaries of your area eg.

(a) No 1 section – aadhabaen 600 PILI JHOPRI naam JHOPRI. Section kebaen wale jawan se JHOPRI tak ki line baen had.



Aadha Dayen



Aadha Bayen

(b) Similarly indicate right boundary.

(c) Dived the ground in to foreground, middle and distance. Having done so start from LEFT to RIGHT systematically and describe.

(d) In attack describe the ground nearest to you first i.e. foreground, then middle and then distance, in defence reverse the procedure.

PART III: INDICATION OF TARGETS

Method of Indication of Easy Targets

10. **Easy Targets**. Can be indicated by the following methods:-

(a) **Indication by Description**. An obvious target can often be described directly. For example, 'No. 1 Section BRIDGE' Here BRIDGE is so obvious that nobody can make a mistake in recognizing it.

(b) **Indication by Direction or Range or Both**. In slightly less obvious cases other aids should be used e.g. direction or range or both. An example of each is given below:-

(i) **Indication by Direction**. No 1 Section BAEN BAGHICHA.

(ii) **Indication by Range**. No 1 Section 600 BAGHICHA.

(iii) **Direction and Range**. When indicting a land mark indicate direction first and then range eg BAEN-600, BAGICHA.



Indication of Difficult Targets

11. The target which cannot be indicated by the methods given above are termed difficult targets. The methods to indicate these are explained in succeeding paragraphs.



The Direction Method

12. This is used to indicate the following:-

(a) The general line of direction, or

(b) A known reference point, or

(c) Another land mark.

13. Unless otherwise stated all direction are taken to be with reference to the general line of direction. The following direction will be used:-

Direction	Measuring
Slight Left / Right	Approximately 10°
Quarter Left / Right	Approximately 22½°
Half Left / Right	Approximately 45°
Three Quarter Left / Right	Approximately 67½°
Full Left / Right	Approximately 90°







PRACTICE

14. After theoretical class, the cadets will be given exposure of subject practically outdoor. This will include identification of targets, as well as indication of easy targets and difficult targets.

CONCLUSION

15. To achieve success in war it is very important that the target is correctly understood and recognized by the troops. By using the methods of indicating the target we can easily indicate and identify the target. The accomplishment of task, can only be achieved after identification of the correct target. Hence adequate training is required to be imparted during peace time.

SUMMARY

- The Correct Sequence of indicating landmarks is Reference Point, Right or Left, Clock, Method, Degree and indication by distance.
- Easy targets can be indicated by Description or Direction or Range or both.
- Different targets can be indicated by Direction method using GLD, a known ref pt or any other landmark.
- Use of Landmarks enables effective description of targets.
- The target of grounds plays vital role in effective identification of landmarks.

ASSESSMENT EXERCISE

Multiple Choice Questions & Fill in the blanks

1. While indicating landmarks, if the ground is to be described, we must start from the _____, go around to the right and finish at the GLD.

(a) Left (b) Farthest
(c) Closest (d) GLD

2. An Object having tactical signification which is indicated with a view to bring down fire on it is called a _____.

(a) Landmark (b) Objective
(c) Target (d) Ground of Tactical Importance

3. What is full form of GLD?

(a) Group Line of Direction (b) General level of Domination
(c) General Level of Distance (d) General Line of Direction

4. During Indication of Landmarks, “Half Left’ corresponds to _____ degrees to your Left.

(a) 23 ½ (b 45
(c) 90 (d) 60

5. An important and unmistakable object, with the help of which you can indicate other landmarks or targets is called a _____.

(a) Important Landmark (b) Reference Point
(c) Man Target (d) Prime Landmark

6. Reference Point is _____

7. Quarter left or right amounts to _____ degrees approximately.

8. “No 1 Section Bridge” is _____ method of describing a target.

9. An object having a technical significance which is indicated with a view to bring down fire on it is called_____.

10. An impartial & unmistakable object with the help of which you can indicate other landmarks is called _____.

Short Answer Questions

Q1. Describe Indication of Landmarks?

Q2. What do you understand by GLD?

Q3. What is direction method?

Q4. Give examples of indication by clock & deg method?

Q5. What is a Reference Point?

Long Answer Questions

Q1. Howmany types of methods are there for indication of targets? Explain it?

Q2. Write a short note on methods of indication of targets?

Q3. Elaborate sequence of description?

Q4. Describe indication of Landmarks?

Q5. How to give General line of direction?

FIELD CRAFT AND BATTLE CRAFT (SD/SW)

CHAPTER FCBC-III: OBSERVATION, CAMOUFLAGE & CONCEALMENT

"If the enemy leaves a door open, you must rush in."



TEACHING INSTRUCTIONS

Period : 03 (120 Min)
Type Year : 2nd Year SD/SW
Conducting Officer : Permanent Instructor

Training Aids: Class Room, Computer with OHP & Screen, Pointer Staff, Presentation, Precis Lesson plan, Board & markers and Open training area for practical classes/ Practice.

Time Plan

- **Introduction : 05 Min**
- **Part I : 15 Min**
- **Part II : 20 Min**
- **Part III : 30 Min**
- **Conclusion : 10 Min**
- **Practical & Training : 40 Min**

INTRODUCTION

1. Camouflage is a type of concealment or a disguise using color, materials or illumination to make an object or organism very hard to detect. Concealment on the other hand is the act of hiding or preventing something or someone from being observed or noticed.

PREVIEW

The lecture will be conducted in following Parts:-

- **Part I: Why things are seen.**
- **Part II: Personal and Equipment camouflage.**
- **Part III: Concealment.**
- **Part IV: Practical Training.**



INTERESTING FACTS

- **Camouflage comes from a French word 'Camoufler'.**
- **Snipers use camouflage and concealment to the Optimum for their own safety as well as maintaining surprises.**
- **Antiradiation paint is used on vehicles and equipment in order to achieve camouflage & concealment.**
- **Drones can be effectively used as a counter measure for camouflage and concealment.**

LEARNING OBJECTIVES

- **Familiarization with Terminologies used and under-standing their definitions.**
- **Sequence of description of Ground.**
- **Various methods used for identification & indication of targets.**

PART I: WHY THINGS ARE SEEN

2. Various factors responsible for things to be seen are as follows:-

(a) **Shape**. Not Just the overall silhouette, beware of the shapes and colors on your suit and whether they match the environment you are in. String ghillies in woodland are as bad as leaf suits in the grass. Attach elastic loops or use elastic bands to add local vegetation to help match yourself to the background.



(b) **Shine**. Anything with a smooth surface, from a plastic buckle to the rifle and pistol will give off a shine. Look closely and a lot of modern camouflages with high polyester content will have a slight sheen to them. Matt spray paint will help but in the case of the rifle, add fabric tape and other materials to hide the surfaces. Spray paint can be used to make excellent patterns and colors but still leave a very uniform flat surface that catches the light. Cover your scope with fabric and cut a hole in to see through (doesn't affect the view) to hide and glint from the lens and also to hide that giveaway black circle.





(c) **Shadow**. Use light carefully, be aware of casting shadows as you move. Move in the shadows if you can. No matter how good the camouflage, you cannot hide that big dark shape of a shadow.





(d) **Silhouette**. The human outline is very distinctive. Break it up, but also be aware of how your shape interacts with the environment. The classic example is “sky lining”, where you move across the top of a hill, outlining yourself. Move lower down, and be aware of the outline you make in the environment. Obviously, the suit will do a lot of the work for you, disguising the shape. If you are running a leaf suit, or ghillie cape or make sure to add 3 Dimensional camo to all to help break up the human outline. Also, the head and shoulders, usually on show the most, must be disguised. Utilize a big hood, and pack extra material on the sides of the head to break up the neck outline.



(e) **Smell**. Humans have a poor sense of smell and so tend to disregard this sense. Wild animals have an extremely keen sense of smell. Many hunters stalk their prey downwind so that the prey doesn’t smell them. Don’t put on aftershave or smelly deodorant. Don’t eat spicy foods and do not smoke beforehand.

(f) **Sound**. Sound is very important on two levels. Obviously, it goes without saying that you need to minimize your own noise by taping anything loose like buckles and straps, making your rifle as quite as possible and moving quietly. When you move and you are fairly close to the enemy, then roll the foot forward on the outside the foot. This will reduce noise from footsteps. The second part of sound is being aware of noise around you.

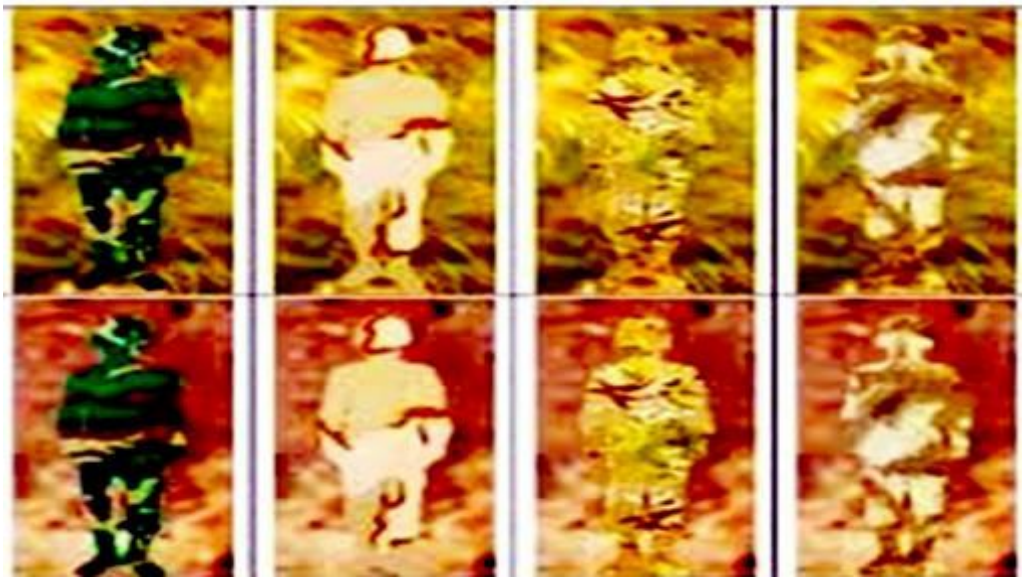


(g) **Speed/ Movement**. Everything you do must be done slowly, unless there is an emergency. Movement is big giveaway; it’s easily spotted by your enemy. Turn your head slowly when scanning an area, change stance slowly, move your limbs slowly, move your weapon slowly and smoothly. Also make sure your suit does not interfere with anything. You don’t want to have to make any excess movement flicking raffia out of your face, or adjusting a hood. Think carefully about what kit you carry too. When selecting a fig, always make sure your load bearing sits high on the body and avoid belt setups.

PART II: PERSONAL AND EQUIPMENT CAMOUFLAGE

3. **Personal Camouflage**. Camouflage means to disguise an object in plain sight in order to conceal it from something or someone. Personal camouflage can be done by following methods:-

(a) Use of Disruptive Pattern.



(b) Clothing and Local Vegetation.



(c) Camouflage of Face.





<u>Camouflage of Equipment</u>.

4. Camouflage of Equipment:-

(a) **<u>Helmet</u>**. Use of Hessian Cloth & Use of Camouflage Net.



(b) **<u>Camouflage of Back Packs</u>**. Packs have square outline which is broken by tying thin ropes on the packs and branches of local vegetation are stuck in the ropes.



(c) **Camouflage of Rifle**. Metal parts of rifles are phosphate coated and hence shine is reduced. Garnish of suitable color should be wrapped over wooden parts / plastic parts to contrast with surroundings.



(d) **Camouflage of LMG**. Wooden/plastic parts of the LMG can be covered with garnish or a combat camouflage cloth in case it is not painted in a camouflage pattern, as shown in the picture below. The LMG trench is camouflaged using a net.



(e) **Camouflage of Dress/Web Equipment**. Equipment like dress binocular, map case, Radio Set should also be camouflaged by breaking its outline and preventing shine, shape on silhouette.





(f) **Fitting of Equipment**. Personal dress and equipment should be packed and fitted properly to avoid any unnecessary noise/tell tale signs.

PART III: CONCEALMENT

5. **Concealment**. If the enemy can see you, he can hit you with his fire. So, you must be concealed from enemy observation and have cover from enemy fire. When the terrain does not provide natural cover and concealment, you must prepare your cover and use natural and manmade materials to camouflage/ conceal yourself, your equipment and your position.

Correct Use of Cover

6. The use of various types of ground and natural cover is very essential to achieve concealment. A cadet however, skillfully camouflaged, not making correct use of cover is likely to be detected. Certain fundamentals for correct use of cover are as follows:-

(a) Whenever possible look through the cover and not over it.

(b) If it is not possible to look through the cover, look round it rather than over it.

(c) It is necessary to look over the cover, avoid breaking straight line.

(d) The sky-line is the worst background you could choose, but if you cannot avoid observing over the cover and against the sky line, use something to break your silhouette.

(e) When firing from inside a building keep well back making use of the shadow.

(f) A rough, dark and irregular background which matches your clothing, provides considerable cover from view.

(g) Isolated cover is dangerous because it will attract attention of the enemy and can be easily indicated in a fire order.

(h) Avoid skyline.

(j) Avoid regulars pacing.

(k) Avoid gaps for fire position.

(l) Cross gaps as a body and at irregular interval at the double.

DID YOU KNOW?

- Camouflage was most commonly used on guns and vehicles.
- Concealment is the most common camouflage technique.
- Each nation developed its unique camouflage pattern after the WW-II.

PRACTICE AND TRAINING

7. After theoretical class, the cadets will be given exposure of subject practically outdoor. The various materials used for camouflage & concealment like Blanco, colors, garnish will be provisioned for the cadets.

CONCLUSION

8. To conclude, the importance of camouflage and concealment can be realized from the following:-

(a) In earlier days it was said "If it can be seen, it can be hit, if it can be hit it can be killed".

(b) But now in the modern warfare "If it can be seen it will be killed".

9. Therefore, camouflage needs greater emphasis and the art of camouflage and concealment reduces the different varieties of soldiers into two main categories viz, "The good and the dead." Concealment is an aid to tactical deception and misinforms the enemy about our intentions and strengths.

SUMMARY

- Objects are seen during day due to their Shape, Shine, Shadow, Surface, Sparing, Smoke, Sound and Movement. This can be coded as 7S
- National resources and vegetation can be effectively used for camouflage of persons as well as equipment
- Various types of cover are Cover from view, cover from fire, avoid breaking of straight line, avoid isolated cover and looking through or around cover

ASSESSMENT EXERCISE

Multiple Choice Questions & Fill in the Blanks

1. In order to camouflage a rifle, it's wooden / plastic parts should be wrapped around with _____ of suitable colour.

(a) Plastic Sheet (b) Garnish
(c) Tape (d) Paint

2. An action of misleading the enemy by concealing or misrepresenting the identity of own troops, equipment, installations and activities is called _____

(a) Tactics (b) Camouflage
(c) Isolation (d) Concealment

3. The following is not responsible for things to be seen during day.

(a) Sound (b) Light
(c) Shine (d) Smoke

4. Cover from view means _____.

5. Things are seen during night due to _____ & _____.

6. _____ is the most common camouflage tech.

7. Concealment is

(a) Concealment from enemy view
(b) Cover from enemy fire
(c) Both (a) and (b)
(d) Engage enemy with fire

8. _____ and _____ are used as a camouflage for a helmet.

9.. _____ and _____ are the two things that are visible at night.

Short Answer Questions

Q1. What is personnel Camouflage?

Q2. Write Factors responsible for things to be seen.

Q3. What all personal equipment will you camouflage before going for any Operation?

Long Answer Questions

Q1. Explain Correct use of Cover.

Q2. Give 3 new innovations for camouflage and concealment.

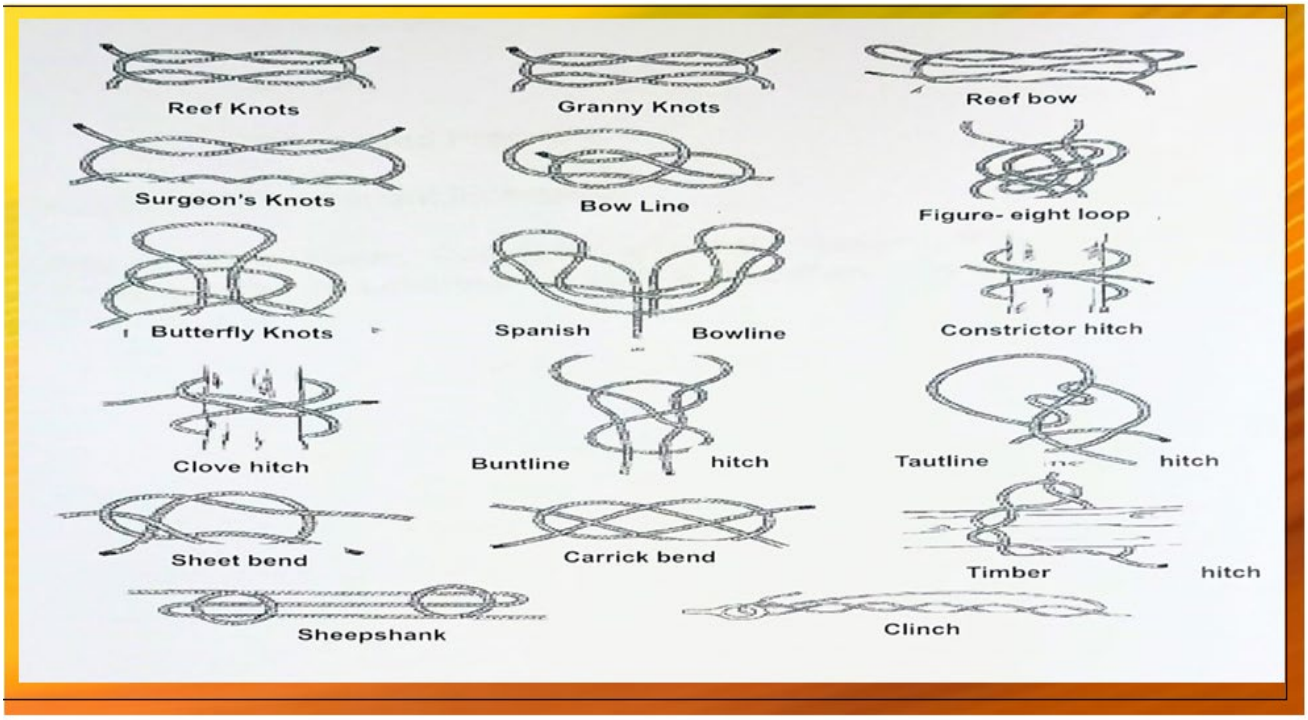
Q3. Elaborate various reasons for objects to be seen during day and night.

Q4. Differentiate between camouflage and Concealment.

Q5. How can drones be used to counter camouflage and concealment.

FIELD CRAFT AND BATTLE CRAFT (SD/SW)

CHAPTER FCBC-IV: KNOTS, LASHING AND STRETCHER



TEACHING INSTRUCTIONS

Period : 02 (80 Mins)
Type Year : 1ST Year (01) & 3rd Year (01) SD/SW
Conducting Officer : Permanent Instructor

Training Aids: Class Room, Computer with OHP & Screen, Pointer Staff, Presentation, Preci Lesson plan, Board & markers and Open training area for practical classes/Practice.

Time Plan

- **Introduction : 02 Mins**
- **Part I : 15 Mins**
- **Part II : 20 Mins**
- **Part III : 40 Mins**
- **Conclusion : 03 Mins**

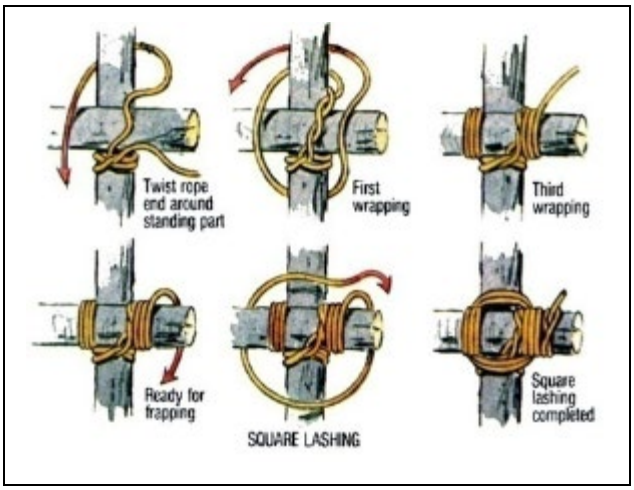
INTRODUCTION

1. The ability to join two pieces of natural material together and so increase their length, gives man the ability to make full use of many natural materials found locally. Knot tying is a useful exercise to obtain better coordination between eyes and fingers.

PREVIEW

The lecture will be conducted in following Parts:-

- **Part I: Types of Knots.**
- **Part II: Lashings.**
- **Part III: Stretchers.**
- **Part IV: Practical Training.**



INTERESTING FACTS

- **The standard size of a stretcher is L - 7' 9" and W - 1' 11" with length of canvas 6'.**
- **The types of Stretchers vary depending upon the nature of use and are different for hospital, ambulance and ship.**
- **Correct Lashing and Knots assist you in river crossing and obstacle crossing.**

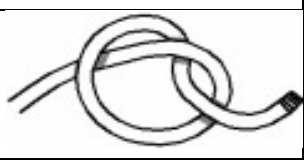
LEARNING OBJECTIVES

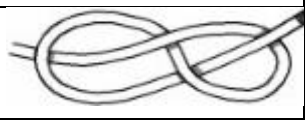
- **Make cadets understand the types & uses of various types of Knots.**
- **Basic understanding of Stretchers and their uses.**
- **Impart on ground training using natural resources.**

PART I: TYPES OF KNOTS

2. A brief description of the use to which the knot may be put is given in this lesson. The diagrams will explain how the knot is tied. The letter "F" means the free or untied end of the rope, and the letter "S" means the standing or secured end.

3. **Knots for Rope ends or for Grips on Thin Rope**.

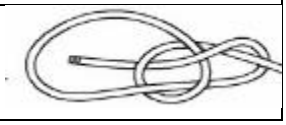
Ser No	Type of Knot	Sample
(a)	**Thumb Knot**. To make a stop on a rope end, to prevent the end from fraying or to stop the rope slipping through a sheave, etc.	

Ser No	Type of Knot	Sample
(b)	**Overhand Knot**. Over hand knot may be put to the same use as the thumb knot. It makes a better grip knot, and is easy to undo.	
(c)	**Figure Eight**. This knot is used as the thumb knot. It is easy to undo, and more ornamental.	

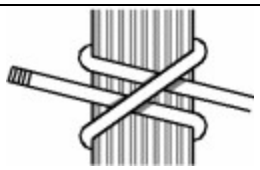
4. **Knots for Joining Ropes**.

Ser No	Type of Knot	Sample
(a)	**Reef Knot**. To securely join two ropes of equal thickness together. Notice the difference in position of the free and standing ends between this and the thief knot.	
(b)	**Thief Knot**. To tie two ropes of equal thickness together so that they will appear to be tied with a reef knot, and will be retied with a true reef knot. This knot was often used by sailors to tie their sea chests, hence the name.	
(c)	**Fisherman's Knot**. For joining two springy materials together; suitable for wire, fishing gut or vines. Two thumb knots (one on each rope) pulled tight. The knots lock together.	

5. **Knots to Make Loops in Rope**.

Ser No	Type of Knot	Sample
(a)	**Bowline**. To form a loop that will not slip on a rope end.	

6. **Knots for Fastening Ropes**.

Ser No	Type of Knot	Sample
(a)	**Clove Hitch**. For securing a rope to as par. This hitch, if pulled taut, will not slip up or down on a smooth surface. A useful start for lashings.	

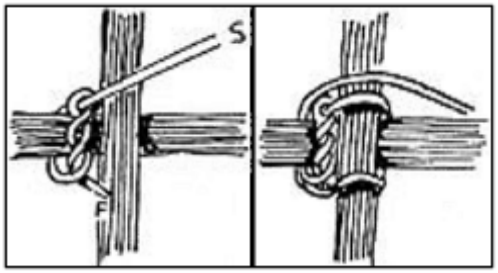
PART II: LASHINGS AND STRETCHERS

Lashings

7. The methods employed to tie with ropes, poles or any rope to a stationary object to securely hold it in place is known as lashing.

(a) **Square Lashing**. To join poles at right angles.

(b) **Frapping Turns**. These are turns that go round the lashing and pull it tight.



8. **Stretchers**. A stretcher, litter, or pram is an apparatus used for moving patients who require medical care. A basic type (cot or litter) must be carried by two or more people. A wheeled stretcher (known as a gurney, trolley, bed or cart) is often equipped with variable height frames, wheels, tracks, or skids.

9. **Basic stretchers**.

(a) Simple stretchers are the most rudimentary type. They are lightweight and portable, made of canvas or other synthetic material suspended between two poles or tubular frame. Many are stored as disaster supplies and are often former military equipment.



(b) The scoop stretcher is used for lifting patients, for instance from the ground on to an ambulance stretcher or on to a spinal board. The two ends of the stretcher can be detached from each other, splitting the stretcher into two longitudinal halves. To load a patient, one or both ends of the stretcher are detached, the halves placed under the patient from either side and fastened back together.

(c) The litter, also known as a rescue basket or Stokes basket, is designed to be used where there are obstacles to movement or other hazards; for example, in confined spaces, on slopes, or in wooded terrain. Typically, it is shaped to accommodate an adult in a face-up position and is used in search and rescue operations. The person is strapped into the basket, making safe evacuation possible. The litter has raised sides and often includes a removable head/torso cover for patient protection. After the person is secured in the litter, it may be wheeled, carried by hand, mounted on an ATV, towed behind skis, a snowmobile, or a horse, lifted or lowered on high-angle ropes, or hoisted by helicopter.

(d) A Reeves Sleeve, SKED, or "flexible stretcher" is a flexible stretcher that is often supported longitudinally by wooden or plastic planks. It is a kind of tarpaulin with handles. It is primarily used to move a patient through confined spaces, e.g., a narrow hallway, or to lift obese patients. Reeves stretchers have six handholds, allowing multiple rescuers to assist extrication.

(e) The Wauck board is also designed for use in small spaces. The patient is secured to the board with straps. It has two wheels and a foldable footrest at one end, allowing the patient to be moved by one person, much as with a hand truck for moving cargo. It can also be used at a variety of angles, making it easier to traverse obstacles, such as tight stairwells.

DID YOU KNOW?

- Stretchers can also be created using natural resources combined with your personal Kit.
- Stretchers can also be effective to control contagious diseases.
- Stretchers can also be prepared using your combat dress and ground sheet.



Stretcher prepared using combat dress



Stretcher prepared using Groundsheet

10. **Wheeled stretchers**. For ambulances, a collapsible wheeled stretcher, or gurney, is a type of stretcher on a variable-height wheeled frame. Normally, an integral lug on the stretcher locks into a sprung latch within the ambulance in order to prevent movement during transport, often referred to as antlers due to their shape. It is usually covered with a disposable sheet and cleaned after each patient in order to prevent the spread of infection. Its key value is to facilitate moving the patient and sheet on to a fixed bed or table on arrival at the emergency department. Both types may have straps to secure the patient.



PART III: PRACTICAL TRAINING

11. After theoretical class, the cadets will be given exposure of subject practically outdoor. Practical classes /Practice will emphasize on the following:-

(a) Making of various types of knots.

(b) Uses of various types of knots.

(c) Preparation & uses of various types stretchers - During the practice it will be ensured that natural resources are used to the optimum.

CONCLUSION

12. Knots, lashings and stretchers are very useful and become very handy for cadets during camps. They can make use of knots for joining or tying 2 to 3 different ropes together and make use of it during rope climbing, rappelling, slithering and other such adventure activities. Similarly, lashings can be used for joining 2 or 3 things together to make a structure that can useful in camps. For evacuation of any injured or causality stretchers are important to be known and used.

SUMMARY

- Various types of knots that can be life saving are:-
  - Thumb Knot
  - Overhand Knot
  - Figure Eight Knot
  - Reef Knot
  - Thief Knot
  - Fisherman’s Knot

- Bowline Knot
- Clove Hitch Knot

➢ Knots come handy while joining ropes and make loops.

➢ Various types of stretchers are:-

- Basic stretcher.
  - Simple stretcher
  - Scoop stretcher
  - The litter (Rescue basket or Stokes basket)
  - Reeves Sleeve, SKED, or "flexible stretcher"
- Wheeled stretcher
- The Nimier stretcher

ASSESSMENT EXERCISE

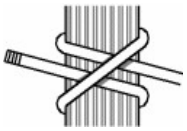
Multiple Choice Questions & Fill in the Blanks

1. Identify the Knot/ Lashing



(a) Square Lashing (b) Thief Knot
(c) Fisherman's Knot (d) Overhand Knot

2. Identify the Knot/ Lashing

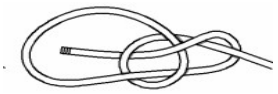


(a) Square Lashing (b) Thief Knot
(c) Fisherman's Knot (d) Clove Hitch

3. Clove Hitch is used for _____.

(a) Join ropes (b) Secure a rope to a Spar
(c) Make loops in rope (d) Make rope-ends

4. Identify the Knot/ Lashing



(a) Square Lashing (b) Thief Knot
(c) Bowline (d) Overhand Knot

5. Letter 'F' means _____end of the rope.

6. Letter 'S' means _____ end of the rope

7. Lashing is of _____ types

8. _____ Stretcher was used by the French Army during WW I.

9. Turns that go round the lashing and pull it tight is called

(a) Frapping Turns (b) Square Lashing
(c) Clove Hitch (d) Bowline

10. _____ is used for joining materials together to a springy fisherman's knot.

11. _____ is used to make a stop on a rope knot.

12. _____ is a type of rope that will not slip on a rope end.

Short Answer Questions

1. How many Types of knots are there? write names.

2. Give various types of Knots for joining ropes.

3. What are types of stretchers?

4. Differentiate thumb Knot & Overhand Knot with appropriate diagram.

5. What is Nimier Stretcher?

Long Answer Questions

1. Define Basic stretchers & Wheeled stretchers. Elucidate your answer with appropriate diagram.

2. Explain various types of Konts & their usages.

FIELD CRAFT AND BATTLE CRAFT (SD/SW)

CHAPTER FCBC-V: FIELD SIGNALS

"Do not repeat the tactics which have gained you one victory, but let your methods be regulated by the infinite variety of circumstances"



TEACHING INSTRUCTIONS

Period : 02 (80 Min)
Type Year : 2nd Year SD/SW
Conducting Officer : Permanent Instructor

Training Aids: Class Room, Computer with OHP & Screen, Pointer Staff, Presentation, Precis Lesson plan, Board & Markers and Open training area for practical classes/.

Time Plan

- **Introduction : 03 Mins**
- **Part I : 05 Mins**
- **Part II : 05 Mins**
- **Part III : 05 Mins**
- **Part IV : 10 Mins**
- **Part V : 10 Mins**
- **Conclusion : 02 Mins**
- **Practical Training : 40 Mins**

INTRODUCTION

1. It is not always possible to give verbal orders through words of mouth because of battle noise, security reasons and intervening distance being too great for effective voice control. Whenever someone wants to convey his message, one has to raise his voice. In older days smoke, sound of drum was the mode to convey messages from one village to another. In Army, different methods are used to convey messages. Today, you will learn one of these methods called Field Signals.

PREVIEW

The lecture will be conducted in following Parts:-

- **Part I: Signals with Hand**
- **Part II: Signals with Weapons**
- **Part III: Signals with Whistle**
- **Part IV: Field Signals as Means of Giving Orders**
- **Part V: Signals by Day & Night**
- **Part VI: Practical Training**



INTERESTING FACTS

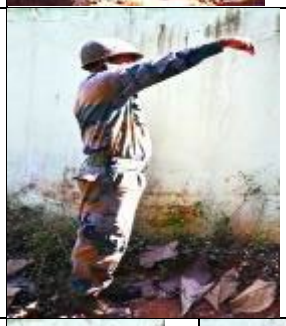
- **The standard size of a stretcher is L - 7' 9" and W - 1' 11" with length of canvas 6'.**
- **The types of Stretchers vary depending upon the nature of use and are different for hospital, ambulance and ship.**
- **Correct Lashing and Knots assist you in river crossing and obstacle crossing.**

LEARNING OBJECTIVES

- **Visual signals include flags, pyro techniques and ground to air signals.**
- **Soldiers can also use other items such as chemical light sticks and flashlights for to signals during inclement weather/ darkness.**
- **Each soldier must pass the signal to the soldier behind him to ensure acknowledgement.**

PART I: SIGNALS WITH HAND

Ser No	Name of Signal	Illustrations	Signal With Hand
(a)	Deploy	Right arm fully extended above head and waved from side to side, palm open.	
(b)	Advance	Right arm swung from rear to front in “under arm blowing” fashion.	 
(c)	Halt	Right arm raised to full extent above head.	
(d)	Turn About	Right arm raised and bent above head.	
(e)	Change Direction	Right arm raised to front in line with shoulder. Body then turned in required direction.	

(f)	Close	Right hand place on top of head, elbow to the right.	
(g)	Follow me	Right arm swung from rear to front above the shoulder in “over arm bowling” fashion.	
(h)	Enemy Approaching	Both hands open, palm in wards at waist level, with inwards scooping motion.	 
(j)	Enemy LMG firing	Right hand thumb down signal.	
(k)	Attack	Punching motion with Right or Left hand according to direction of attack.	 
(l)	Closed to Rendezvous	Close sign followed by both hands clasped in front of body at waist level.	 

(m)	Infantry obstacle ahead	Both hands crossed in front of body at the waist, palm open downwards.	

DID YOU KNOW?

- Field signals are alternative to voice control.
- Each soldier must pass the signal to the soldier behind him to ensure acknowledgement.

PART II: SIGNALS WITH WEAPONS

Ser No	Name of Signal	Signals with Weapons	Illustrations
(a)	Enemy in Sight	Rifle held above the head parallel to the small number muzzle in the direction of the enemy.	
(b)	Enemy in Sight in large number	As per (a) above, but arm moved up and several times.	 
(c)	Advance	Both arms raised to form the letter U	

PART III: SIGNALS WITH WHISTLE

Ser	Name of Signal	Signals With Whistle
(a)	Cautionary Blast	A short blast to draw attention to assign all or order about to be given.

Ser	Name of Signal	Signals With Whistle
(b)	The Alarm Blast	A succession of alternate long and short whistle blasts.
(c)	Enemy Aircraft	A succession of short blasts.
(d)	Enemy Aircraft departed	Two long blasts repeated at interval of five seconds.

PART IV: FIELD SIGNALS AS MEANS OF GIVING ORDERS

2. Field signals are alternate means of giving orders and to control troops when voice control is not possible. Control over troops deployed can be done better by field signals than by voice control. There are various occasions when voice control is not possible. They are:-

(a) **Battle Noises**. In war, due to firing and vehicle movement very high noise will be produced, so it may not be possible for a Commander to give voice message.

(b) **Need for Silence**. There are certain operations which, by design are carried out in utmost silence, e.g.:-

(i) Ambush.

(ii) Patrolling.

(iii) Raid.

(iv) Cordon.

(c) **Intervening Distances are Too Large**. When Infantry takes the battle field they come across many obstacles in ground such as the ditch, rivers, and mountains. Under such circumstances we will have the only choice left with us is field signals for communication.

3. **Methods to Attract Attention Of Troops**. Before any field signal is executed, the commander has to attract the attention of troops. Methods to attract attention of troops are:-

(a) A Short Blast of Whistle.

(b) A Bird Call.

(c) Whistle by Mouth.

(d) Clicks (By using tongue).

(e) Clicks by Fingers.

4. **Other Methods of Communication**. Besides field signals there are various other means of communication in the Army which are as follows:-

(a) Dispatch Runners.



(b) Dispatch Rider.



(c) Radio Sets.

PART V: FIELD SIGNALS DAY & NIGHT

5. **Field Signals By Day**. Some of the visual signals used during the day are flags and mercury coated mirrors. Flags are very effectively used by Navy on board a ship by following ways:-

(a) Flags: (Red, Green and White flags).

(b) Mercury coated mirrors.

(c) Smoke.

(d) Miscellaneous: Various signals can be improvised and pre-arranged as under:-

(i) Clothes superficially hung out to dry.

(ii) Hurricane lamp, kept in the window.

(iii) Flashing of torch is used as Morse Code.

(iv) Applying various cooler/ signs on forehead and arms.

6. **Field Signals by Night**. Some of the field signals that can be used at night are:-

(a) Pre decided signals on a walkie talkie.

(b) Click by fingers.

(c) Clicks by using tongue.

(d) Whistle by the mouth.

(e) Use of rope.

(f) Use of color light.

(g) Use of blacked out torch

(h) Firing of weapon

PART VI: PRACTICAL TRAINING

7. After theoretical class, the cadets will be given exposure of subject practically outdoor. Practice will include the following:-

(a) Field Signals with hand.

(b) Field Signals with weapons.

(c) Field Signals with whistle.

(d) Field Signals as means of giving orders.

(e) Field Signals during day & night.

CONCLUSION

8. The fighting efficiency of a unit/sub unit depends on sound communication system which helps commanders at all levels to exercise command and control effectively. Every commander must influence the battle by his personal touch which his achieved by good signal communications. Field signals become a part of movement of a good section/platoon commander. At section/platoon level, the commander directly influences the battle by the use of field signals. Victory in battle will come to that section/ platoon whose men are familiar in the use of field signals

SUMMARY

- When it is practically not possible to use verbal control orders, field signals are used to communicate with troops.
- Field signals can be given using hand, weapon or whistle.
- Various methods can also be improvised to communicate with troops during inclement weather as well as during works at darkness.

ASSESSMENT EXERCISE

Multiple Choice Questions and Fill in the Blanks

Q1. "Right Hand Thumb Down' is a Field Signal for _____.

(a) Enemy in Sight (b) Turn about
(c) Enemy LMG firing (d) Close

Q2. Field signals are alternate means of giving orders and to control troops when _____ control is not possible.

(a) Tactical (b) Voice
(c) Military (d) Line of Sight

Q3. "Right arm raised to full above head' is a Field Signal for _____.

(a) Advance (b) Attack
(c) Halt (d) Close

Q4. There are certain operations which, by design are carried out in utmost silence. Which are these?

(a) Ambush (b) Cardon
(c) Patrolling (d) All of these

Q5. Air photographs allow gradient to be seen in relief with the help of a _____.

(a) Binocular (b) Stereoscope
(c) Periscope (d) Stethoscope

Q6. _____ & _____ are the occasions when voice control is not possible.

Q7. _____ & _____ can be two improvised signals.

Q8. _____ & _____ can be two Field signals used at night.

Q9. Whistling in succession of short blasts is a signal for _____.

(a) Cautionary Blast (b) Alarm Blast
(c) Enemy Aircraft (d) Enemy Aircraft Depart

Q10. Rifle held above the head parallel to the small number muzzle in the direction of the enemy is signal for:-

(a) Enemy in sight
(b) Enemy in sight in large number
(c) Advance
(d) Deploy

Short Answer Questions

Q1. Explain Field signals both by Day & Night?

Q2. What are various methods of communication?

Q3. What are various methods to attract attention of Troops?

Q4. Explain two signals with whistle?

Q5. Explain Signal for Enemy in Sight?

Long Answer Questions

Q1. Analyze field signals as means of giving orders.

Q2. Explain various types of signals with hand.

Q3. Describe various signals with weapon & whistle.

Q4. Explains various signals with whistle?

Q5. Explain following signals?

(a) Deploy
(b) Enemy approaching
(c) Attack

FIELD CRAFT AND BATTLE CRAFT (SD/SW)

CHAPTER FCBC-VI: FIRE AND MOVE CAPSULE

"If it is your advantage move forward, if not, stay where you are."



TEACHING INSTRUCTIONS

Period	**:**	**03 (120 Min)**
Type Year	**:**	**3rd Year SD/SW**
Conducting Officer	**:**	**Permanent Instructor**

Training Aids: Class Room, Computer with OHP & Screen, Pointer Staff, Presentation, Precis Lesson plan, Board & Markers and Open training area for practical classes.

Time Plan

➢ **Introduction**	**:**	**05 Mins**
➢ **Part I**	**:**	**10 Mins**
➢ **Part II**	**:**	**10 Mins**
➢ **Part III**	**:**	**10 Mins**
➢ **Part IV**	**:**	**10 Mins**
➢ **Part V**	**:**	**15 Mins**
➢ **Part VI**	**:**	**10 Mins**
➢ **Part VII**	**:**	**10 Mins**
➢ **Conclusion**	**:**	**05 Mins**
➢ **Practical Training**	**:**	**40 Mins**

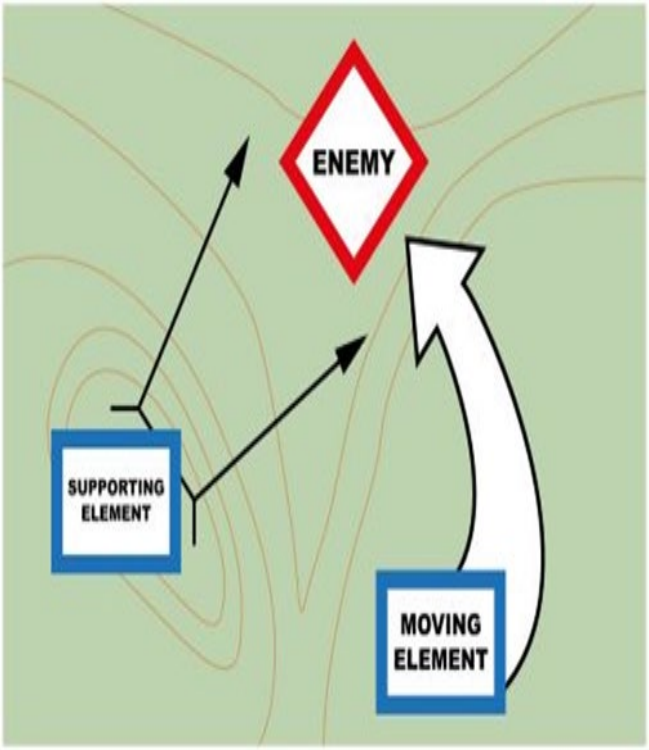
INTRODUCTION

1. The primary aim of infantry is to close in with the enemy & destroy him. The aim of getting close is achieved by making skillful use of ground. A clever enemy will however, deny to use of such ground which you may need. When such cover is denied by the enemy, we may have to move in open. Once we are forced to move in open, a part of our force will have to fire on enemy position & force him to keep his head down. This would render the enemy incapable of bringing down aimed fire at us while we are on the move. This process of keeping one element on the ground to give covering fire, while the other element is on move, is called Fire & Move. To understand the fire and move capsule cadets have to understand basic nuances.

PREVIEW

The lecture will be conducted in following Parts:-

- **Part I:Important Terms**
- **Part II: When to Use Fire and Move**
- **Part III: Basic Consideration of Fire & Move**
- **Part: IV-Use of Ground & Common Mistakes**
- **Part V: Selection of Fire Positions**
- **Part VI: Movement when under Fire**
- **Part VII: Crawls**
- **Part VIII: Practical Training**



INTERESTING FACTS

- **Fire orders are standardized verbal instructions given by a commander to control rate, direction and type of fire.**
- **Type of fire depends upon enemy's movement.**

LEARNING OBJECTIVES

- **Explain the requirement of fire & move.**
- **Basic understanding of various fire position & Equipment.**
- **Impart practical training based on classes conducted.**

PART I: IMPORTANT TERMS

2. Certain terms given below are commonly used in connection with fire control:-

(a) **Fire Unit**. Any number of men firing under the command of one man, usually, a section. The person responsible for giving the executive order of fire, is the fire unit commander.

(b) **Fire Direction Orders**. These are the orders which the fire unit commander receives from his superior, telling him when, at what target and with what intensity to open fire. A section commander will receive fire direction orders from his platoon commander. They include special directions about opening and withholding fire.

(c) **Fire Control Orders**. These are orders given by the fire unit commander to direct and control the fire of the section. Emphasis should be on control and supervision. These orders are the final with complete instructions after all factors have been considered and before fire is actually opened.



(d) **Arc of Fire**. This denotes the area of ground for which the fire unit is responsible and within which it will engage targets. An arc of fire, must not be confused with field of fire which is the area upon which it will be possible to fire effectively in any direction.

PART II: WHEN TO USE FIRE & MOVE TACTICS

3. Fire & Move tactics are used in following circumstances:-

(a) The enemy has opened SA fire which is effective.

(b) When own troops have seen the enemy first - within 400 to 700 meters.

(b) When the enemy is known or suspected to be in a certain area, then fire & movement tactics may be adopted when the troops reach within the effective range of enemy weapons/ observation. (Instructor to explain as to what could be the effective range of enemy's personal weapon).

(c) To cross obstacles by day or by night, e.g. nullahs /rivers.



PART III: BASIC CONSIDERATION OF FIRE AND MOVE

4. The primary aim of the infantry is to close with the enemy and destroy him. It is only possible to move forward against opposition by skillful use of ground, with the help of supporting fire or by a combination of both. The enemy will select positions which, as far as possible, give no ground cover to the attackers. By means of fire, mines and other obstacles he will attempt to halt the latter's advance. Supporting fire is, therefore, necessary to keep the enemy's head down and make movement possible. The combination of fire and movement is the basis of platoon and section tactics. It demands from the soldier the highest standards of weapon training and field craft.

5. Five basic considerations for fire and movement are as under:-

(a) There should be no exposed ground without covering fire.

(b) Control by the commander.

(c) The angle of covering fire from direct firing weapons should be as wide as possible without loss of control or time.

(d) **Miscellaneous**. Any identifiable visual or audio signal which is pre-arranged between the sender and receiver can be used as a signal. Some examples are:-

(i) Whistle.

(ii) Hurricane.

(iii) Flashing.

(iv) Replication of various types of natural saint like the sounds that the animals make.

(e) Sounds that the animals make.

Types of Control Orders

6. There are four types of control orders. These are explained below:-

(a) **Delayed Fire Order**

(i) "No. 1 Section –600-enemy approaching await my orders".

(ii) "No.1 Section 600-enemy advancing through jungle, fire when enemy reaches, open ground".

(b) **Opportunity Fire Order**. "No.1Section. Enemy hidden in broken ground, fire when seen".

(c) **LMG Group Fire Order**. "LMG group sight down–enemy running left or right– fire".

(d) **Full Fire Order**

(i) **Pin Point Target**. "No 2 Rifleman–300 Tree Right– 50 REDHUT-enemy sniper-fire".

(ii) **Area Target**. "LMG-Group-500-Tree-Right Grove, enemy section, burst fire".

Sequence of Fire Order

7. An accepted sequence should always be used in order to avoid confusion, or misunderstanding. The suggested code work is 'GRIT', each letter of which signifies stage in fire orders as follows:-

G - The Group of the section which is addressed, that is the LMG group or the whole section. An order starting with "No 1 Section" indicates that the whole of No.1 Section will fire. "LMG-Group or Rifles Group" means that the group named only will fire.

R - The Range to the target. To ensure accuracy of fire and to concentrate attention on a limited area of ground.

I - The Indication of the point of aim by its description.

T - The type of fire to be employed.

PART III: BASIC CONSIDERATION OF FIRE AND MOVE

8. The primary aim of the infantry is to close with the enemy and destroy him. It is only possible to move forward against opposition by skillful use of ground, with the help of supporting fire or by a combination of both. The enemy will select positions which, as far as possible, give no ground cover to the attackers. By means of fire, mines and other obstacles he will attempt to halt the latter's advance. Supporting fire is, therefore, necessary to keep the enemy's head down and make movement possible. The combination of fire and movement is the basis of platoon and section tactics. It demands from the soldier the highest standards of weapon training and field craft.

9. There are five basic considerations for fire and movement. These are as follows:-

(d) There should be no exposed ground without covering fire.

(e) Control by the commander.

(f) The angle of covering fire from direct firing weapons should be as wide as possible without loss of control or time.

(g) Full use should be made of all available cover. When cover is lacking the use of smoke (smoke grenade) should be considered.

(h) Full use should be made of all available weapons for covering fire.

PART IV: APPRECIATION OF GROUND & COVER AND TYPES OF COVER

Ground Appreciation

10. In a battle, fire and movement is applied according to the type of country over which it is fought. In open country the problem is how to find cover; in close country, there is the difficulty of finding positions with good observation and field of fire. In attack or defence, the skillful use of ground can help to gain in developing an eye for ground. Ground should be considered from the enemy's point of view. It should be appreciated under the following headings:-

(a) Fire positions.

(b) Observation positions.

(c) Cover from fire.

(d) Cover from view.

(e) Obstacles.

11. **Types of Cover**. Cover from view is often not cover from fire, especially if the move to cover has been seen by the enemy. Concealment from enemy air and ground observation is the chief means of gaining surprise. Some of the main types of cover are:-

(a) Undulating ground which is the least obvious form of cover; when skillfully used, it protects from direct fire and gives no ranging marks to the enemy.

(b) Sunken roads, beds of streams and ditches which give good cover from view and often from fire as well. However, there is always a danger that the enemy may pay special attention to them; they may be mined or booby-trapped and precautions against ambush must be taken. If the roads or ditches are straight, the enemy will be able to fire down them in enfilade.

(c) Hedges and bushes give cover from view but not from fire. In open country they may make good ranging marks for the enemy.

(d) Standing crops give cover from view but movement through them can generally be detected.

(e) Woods which give cover to men and vehicles from enemy air and ground observation. They give some protection from small arm fire but HE bombs and shells will explode in the branches of trees and will cause heavy casualties unless troops are dug in and have overhead protection.

(f) Buildings and walls afford concealment and protection from small arms fire and shell splinters. When isolated they make good ranging marks for the enemy.

PART V: USE OF GROUND FOR MOVE AND COMMON MISTAKES

12. **Dead Ground**. Ground which a soldier cannot see from his position is called dead ground. Platoon and section commanders should be able to recognize ground which is likely to be dead to the enemy. Ground can only be described as dead in relation to the position of an observer. Troops under cover or in dead ground are safe from enemy observed fire but not from in direct fire. These are as are always likely to be selected by the enemy as defensive fire tasks for his artillery and mortars. Dead ground is also safe from detection by battle field surveillance radars, as these have line of sight limitations.

13. **Common Mistakes**. The wrong use of ground may lead to casualties and loss of surprise; some common mistakes are:-

(a) Carelessness by troops while making a reconnaissance, such as unfolding a map in the open or not using a covered approach to an OP.

(b) Unnecessary movement in a position over looked by the enemy.

(c) Using conspicuous landmarks such as isolated trees, bushes or cottages.

(d) Halting troops near road or track junctions or other mapped features which are always registered as targets by the enemy.

(e) Bad track discipline.

(f) Failure to guard against enemy air observation.

Selection of Fire Positions

14. The ideal fire position should be:-

(a) Provide cover from fire.
(b) Provide cover from view.
(c) Afford a good view of the ground to be watched or target to be engaged.
(d) Provide room in which to use the weapon freely.
(e) Have a covered approach.
(f) Be easy to advance from.

15. The selection of fire positions requires a knowledge both of the characteristics of weapons and of the use of ground. A direct firing weapon must be sited with an eye at the level from which it is to fire. A target which is clear to a man standing may be invisible to one lying down.

16. Sometimes it may be necessary to site fire positions on trees, roof tops, or walls to produce fire effect. This may result in plunging fire, but this must be overcome by accurate shooting. Cunning concealed fire positions will puzzle the enemy, protect the troops from observed fire and safeguard them against air attack.

Fire Control in Attack and Defence

17. There is a big distinction between fire control in attack and in defence. In attack men should be allowed a great deal of latitude in opening fire. Speed and immediate fire effect is what is required. With a well concealed enemy it will often be necessary to “neutralize” an area by fire since few definite targets will be visible. In defence, the vital factor in fire control is that early opening of fire may give away positions to the enemy and jeopardize concealment. Normally, a section commander will lay down a line in front of his section post beyond which fire will not be opened without his orders. This is particularly important where a long field of fire is available. In any case fire will normally be opened on the orders of the section commander.

PART VI: MOVEMENT WHEN UNDER FIRE

18. Movement in the face of the enemy should be covered by fire. This does not mean that it is impossible to move unless a heavy weight of fire is brought down on the enemy. An important part of an attack is the movement towards the objective, supporting fire is one of the aids to that movement. A knowledge of how to move and how to use ground for movement is essential to enable troops to close with the enemy with minimum casualties, undetected in the zone of arc of battle field surveillance radars.

19. Usually, troops advancing by day in action will move at a brisk walking pace until they make contact; in the final stages of the assault, they will double. They may have to double or crawl at other times; for example, if attacking troops move into enemy defensive fire, it is usually best to double forward and through it; to lie down is often dangerous as well as useless. Doubling and crawling are both tiring however, and should only be used in short spells in critical situations particularly for crossing open ground in full view of the enemy. The

commander must himself decide on his pace from his personal knowledge of the state of fitness of his men. In general, the aim must always be to keep moving determinedly towards the enemy at the best possible speed.

20. When crossing an open space like a gap in a hedge, it is best for the whole section or group to double across it together, before the enemy has time to fire effectively. When wider gaps are under enemy observation, it may be necessary to filter men across now and again by crawling in ones and twos.

PART VII: CRAWLS

21. A knowledge of how to move correctly and how to use ground is important to enable individuals and groups of individuals to close with the enemy, while exposing themselves as little as possible to enemy's view and fire. A knowledge to move correctly using correct cover as per the ground is essential. This is also essential for snipers and the patrols to accomplish their tasks. Every feature, natural and artificial, must be used to provide cover from the fire and view.

22. The method of movement is given below:-

(a) **Crawling by Day without Rifle**.

(i) Monkey Run.
(ii) Leopard Crawl.
(iii) Roll.

(b) **Crawling by Day with Rifle**.

(i) Leopard Crawl.
(ii) Side Crawl.

(c) **Crawling by Night**.

(i) The Ghost Walk.
(ii) The Cat Walk.

CRAWLING BY DAY

23. **Monkey Run**. There is useful for moving behind low cover up to about two feet high. Go fast for about 15 meters, drop flat, pause and then crawl further. The method of crawling is explained below:-

(a) **Without Arms**. This is done on hands and knees. Hands are moved forward in turn and so are the knees. Every time a hand is moved forward, the knee corresponding to it is also moved forward to the spot vacated by the hand. When silence is more important than speed. The knee should be placed on the exact spot vacated by the hand. When silence is more important than speed, the knee should be placed on the exact spot vacated by the corresponding hands. The hand should reach forward to safe place.

(b) **With Rifle**. The rifle may be held in the Right hand at the point of balance. Care must be taken to keep the muzzle up to prevent dirt from getting in.

24. **Leopard or Stomach Crawl**. This is useful for moving behind low cover. The method of crawling is explained below:-

(a) **Without Arms**. This is done on elbows and knees. Hug the ground with chest and crouch flat on the ground and arms, outstretched in front. Propulsion is obtained from alternate elbows and knees movement. While crawling roll slightly from side to side as each knee is bent. Avoid kicking up the heels and keep the head, body and elbows close to the ground. Alternatively, one knee only can be used the other led trailing along the ground.

(b) **With Rifle**. This is done on knees and elbows by moving the rifle forward holding the RIGHT hand under the rifle between the points of balance and the outer hand. An alternative method is to grip the rifle diagonally across the body with small of the butt underneath the RIGHT armpit.

25. **Walk**

(a) Never walk with the head bent as this affords no protection but hinders observation. Walk with head up and observe all the while. Silence is important when crossing hard ground, the least noise is made if the outside of the sole of the boot is placed first on the ground. Balance is essential to silent movement and this is assisted by keeping the knees slightly bent.

(b) **With Rifle**. Rifle is held in the LEFT hand across the body, ready for instant action. Carry the rifle so that it looks as a part of you. The alternative method is to carry rifle near the front swivel and the RIGHT hand holding the small of the butt.

26. **Roll**

(a) Roll away keeping the arms to the sides or stretched forward. This is often the quickest way of getting away from a spot where the enemy has seen you. Practice is necessary if the tendency to giddiness is to be overcome.

(b) **With Rifle**. When rolling to the right keep the rifle into the RIGHT side and vice versa.

Crawling with LMG

27. **Knee Crawl (No 1)**

(a) This is useful for moving behind waist height cover e.g. wall, hedgerows.

(b) This one is with the LMG only. The LMG is held in the left HAND. LEFT foot, Right knee and Right hand are placed on the ground. The bodyweight is taken on the RIGHT knee. The Right and the Left foot are moved forward simultaneously followed by the RIGHT knee.

28. **Side Crawl (No 1)**. Lie on a side and rest the gun on the instep of lower leg which is kept flat to the ground. Hold the carrying handle with the Right hand and the barrel with the LEFT. Propulsion is obtained by kicking with upper leg.

29. **Leopard Crawl (No 1)**. Take the gun forward to the limit of the extent of the arms and propel the body forward by legs using the leopard crawl method.

DID YOU KNOW?
➢ A very effective method in open ground is “Crack and Thump”.
➢ Situation awareness is a MUST.

30. **Combined Leopard Crawl (Nos 1 & 2)**. No1 grip the small of the butt in her left hand, No 2 grips the bipod legs in her Right hand. They both move forward using the leopard crawl. No 1’s LEFT arm` and leg keep step with No. 2’s RIGHT arm and leg and vice versa.

THE LEOPARD CRAWL (A)



THE LEOPARD CRAWL (B)



31. **Combined Leopard Crawl by Bounds (Nos 1 & 2)**. Nos 1&2 crawl forward 2 to 3 meters, and lift the gun forward. In this method the gun is kept continuously in the firing position.

CRAWLING BY NIGHT

32. **Ghost Walk**. For all night movements silence is more important than speed. Silence can only be achieved by perfect balance. Stand up, lift the legs high to avoid long grass and sweep them outwards in a semicircular motion. Feel gently with the toes for a foot hold. Makes sure that one foot is safe before the next foot moves, and knees lightly bent. Always lie down when you halt at night.

33. **Cat Walk**. Get down on the hands and knee and move each hand forward searching the ground carefully, making sure there are no twigs, then raise the knee and put it down on the spot where the hand is. Then move the hand forward again. This is very slow method but very sure.

34. **Kitten Crawl**. If the ground is covered with twigs the normal stomach crawl make noise. When moving very close to the enemy and when perfect silence is essential the only sure method is to keep raising the whole body off the ground on the fore arms and the toes

pressing forwards lowering the body, feeling carefully with hands each time. This is very slow and tiring method which requires considerable practice, but is invaluable. Accurate information at night can often only be obtained by movement very close to the enemy.

35. **Stalk**. Ground however carefully selected together with camouflage alone is not sufficient to enable us approach the enemy. One must use it like an animal staking its prey instinctively and unerringly. The use of ground and camouflage combined with the actual move forms the basis of all minor tactics.

PART VIII: PRACTICAL TRAINING

36. After theoretical class, the cadets will be given exposure of subject practically outdoor. This will include the following:-

(a) How to give various fire control orders.

(b) Selection of on ground positions.

(c) How to do movements in face of the enemy.

(d) Various types of crawls with and without weapon.

CONCLUSION

37. Fire & Move is the basic skill of tactics. Its training provided to all the officers and Jawans. The basic aim is to enable one detachment of soldiers to move while the enemy is pinned down by the effective fore of the other temporarily static body soldiers. It is the most effective and safest method to move during operations or when in contact with the enemy.

SUMMARY

- Various circumstances under which fire & movement tactics may be used are when enemy has opened effective fire, when troops have seen within 400-700 meters or to cross obstacles by day & night.
- Selection of fire position & sticking to fire control is of paramount importance during any operations.
- The sequence of fire Control Order is GRIT.

ASSESSMENT EXERCISE

Multiple Choice Questions & Fill in the Blanks

Q1. In Fire and Move tactics, the angle of _____ should be as wide as possible without loss of control or time.

(a) Direct Fire (b) Indirect Fire
(c) Enemy Fire (d) Covering Fire

Q2. Orders given when there is no time to give out a Full fire order, are called _____ Fire Control Orders.

(a) Delayed (b) Opportunity
(c) Brief (d) Full

Q3. Orders given by the fire unit commander to direct and control the fire of his fire unit are called _____.

(i) Fire Direction Order (ii) Fire Control Order
(iii) Attack Order (iv) Firing Orders

Q4. The area of ground for which the fire unit is responsible and within which it will engage targets is called_____.

(a) Arc of Fire (b) Firing Range
(c) Field of Fire (d) Firing Area

Q5. Orders given when there is notmoving continuously seen by everyone in the section, are called _____ Fire Control Orders.

(a) Opportunity (b) Delayed
(c) Brief (d) Full

Q6. GRIT stands for _____, _____, _____ & _____.

Q7. The area of ground for which a fire unit is responsible is called _____.

Q8. The Orders that the fire unit commander receives from his superior is called _____.

Q9. The basic principle of fire & move is _____.

Q10. Getting down on hands & knees, and moving each hand forward searching the ground carefully is called _____.

(a) Kitten Crawl
(b) Cat Walk
(c) Leopard Crawl
(d) Ghost Walk

Q11. The quickest way of getting away from a spot where the enemy has seen you is called

(a) Knee Crawl (b) Side Crawl
(b) Kitten Crawl (d) Roll

Q12. Orders given by Fire Unit Commander to direct and control the fire of the Section is called _____.

Short Answer Questions

Q1. Write five basic considerations for fire and movement.

Q2. Write Ideal fire positions.

Q3. What are the factors for selecting a fire position?

Q4. What do you understand by fire direction order?

Long Answer Questions

Q1. Explain all types of Crawling.

Q2. Describe Types of Cover.

Q3. Elaborate sequence of fire control order.

Q4. What are the points for section Commander and what are the methods of giving the orders.

FIELD CRAFT AND BATTLE CRAFT (SD/SW)

CHAPTER FCBC VII: SECTION FORMATIONS (FCBC-VII)

"Let your rapidity be that of the wind your compactness that of the forest."



TEACHING INSTRUCTIONS

Period	:	**01 (40 Min)**
Type Year	:	**3rd Year SD/SW**
Conducting Officer	:	**Permanent Instructor**

Training Aids: Class Room, A Computer with OHP & Screen, Pointer Staff, Presentation, Preci Outdoor Trg in Trg Area, board & markers and Open training area for practical classes.

Time Plan

➢ **Introduction**	:	**05 Mins**
➢ **Part I**	:	**05 Mins**
➢ **Part II**	:	**05 Mins**
➢ **Part III**	:	**05 Mins**
➢ **Part IV**	:	**05 Mins**
➢ **Part V**	:	**05 Mins**
➢ **Conclusion**	:	**10 Mins**

INTRODUCTION

1. Various formations are used when troops come in contact with the enemy and the type of formation adopted is entirely dependent on the following four basic factors:-

(a) Degree of control required to be exercised by the Section Commander.

(b) Type of ground.

(c) Necessity of bringing down maximum fire with minimum delay.

(d) Task.

PREVIEW

The lecture will be conducted in following Parts:-

- **Part I: Types of formations**.
- **Part II: Section Battle Drill**.
- **Part III: Scout**.
- **Part IV: Types of Patrols**.
- **Part V: Practical Training**.



INTERESTING FACTS

- **Patrolling without being detected is of paramount importance**.
- **Section Formation vary based on the nature of ground and weather conditions**.
- **Patrolling during day and night have significant differences**.

LEARNING OBJECTIVES

- **Explain Various Types of Formations by a Section**.
- **Role, meaning and tasks of scouts**.
- **Explain about various types of Patrols**.

PART I: TYPES OF SECTION FORMATIONS (EXPLANATION WITH DIAGRAM)

2. A Section is the smallest sub unit of an Infantry Battalion and is capable of undertaking independent task. It consists of ten persons who are organized in Rifle group and Support Group. Different types of formations adopted by a section are as under with its advantage & disadvantages:-

SINGLE FILE FORMATION

(a) Single file formation is a formation that is useful for control, moving along narrow paths, and following linear features.

(b) In this formation, each person follows the person in front of them.



ADVANTAGE	DISADVANTAGE
Good for control, not vulnerable to enfilade fire, useful for moving along ditches, narrow defiles and so on.	Bad for fire production, vulnerable to frontal fire.

FILE FORMATION

(a) File formation is a formation where troops are lined up in a column, one behind the other. The width of a column of troops is measured by the number of files.

(b) File formations are useful when troops are unsure of the enemy's position because the overlapping fields of fire from each soldier provides protection from a potential flanking attack.



ADVANTAGE	DISADVANTAGE
Good for control. Useful for moving along broad roads, wide nullahs and so on.	Not good for fire production, vulnerable to frontal fire.

ARROW HEAD FORMATION

An arrowhead formation is a tactical formation used to move across open ground while providing security to the front and rear. It is also known as a "**V**" shape formation.



ADVANTAGE	DISADVANTAGE
Good depth, not vulnerable to frontal fire, good for fire production, probably the best formation for crossing open ground.	Control more difficult than in diamond.

SPEAR HEAD FORMATION

A spearhead formation is a formation where soldiers are organized to maximize their offensive and defensive capabilities.



ADVANTAGE	DISADVANTAGE
Good depth, less vulnerable to enfilade fire than arrow head, LMG group not committed immediately on contact.	Control difficult. Delay in fire production.

DIAMOND FORMATION

The diamond formation is a military formation used in warfare, especially by cavalry, throughout history. It's used when crossing open country at night and is easy to control, while providing good protection and observation.



ADVANTAGE	DISADVANTAGE
Good for control, not vulnerable to enfilade fire, good for all round fire production and observation.	Present a good target to frontal fire. Not very good for fire production to the front.

EXTENDED LINE FORMATION

An Extended line formation is a battle formation where soldiers are positioned in ranks, with each rank about half a meter apart and soldiers in each rank close together a line formation is a battle formation where soldiers are positioned in ranks, with each rank about half a meter apart and soldiers in each rank close together.



ADVANTAGE	DISADVANTAGE
The formation used in the final assault. Very good for fire production (from the hip) and bayonet fighting.	Control difficult. Very vulnerable to enfilade fire, no depth.